

Universal Station^X System Administration

**Release 300
UX11-403**

Universal Station^X

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System Administration**

Release 300

UX11-403

11/99

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About This Publication

Purpose

This publication is provided to aid configuration/operation of Universal Station^X Release 300 hardware/software.

Use this publication to install, configure, and operate the Universal Station^X. If you are unfamiliar with general LAN (Local Area Network) configuration techniques or details of the specific LAN configuration to which the station is attached, your Local Area Network Administrator will be needed to assist in configuration/troubleshooting.

Revision Notes

The following list provides notes concerning the current revision of this document.

Doc ID	Rev. No.	Date	Notes
UX11403	2	11/08/99	Changes made to Revision 2 of this document have been marked with change bars in the outside margin. Overview of Changes 1. Additional steps added to the menu option backup procedure. 2. Additional steps added to the menu option restore procedure. 3. A new section, Change Logical Volume Sizes, added.

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Publication Title	Publication Number	Binder Title	Binder Number
<i>Universal Station^X Customer Release Guide</i>	UX04-493	Universal Station ^X	3095
<i>Universal Station^X System Administration Manual</i>	UX11-403	Universal Station ^X	3095
<i>Universal Station^X (Ergonomic) Service</i>	UX13-530	Universal Station ^X	3095
<i>Universal Station^X Service</i>	UX13-510	Universal Station ^X	3095
<i>Five/Ten-Slot Module Service</i>	LC13-500	LCN Service - 2	3060-2
<i>Hardware Verification Test System HVTs-Universal Station^X</i>	SW13-511	LCN Service - 3	3060-3
<i>U^XS Specification and Technical Data</i>	UX03-503	System Summary-2	3010-2
<i>System Site Planning</i>	SW02-550	System Site Planning-1	3020-1
<i>LCN Guidelines</i>	LC09-510	LCN Installation	3025
<i>LCN System Installation</i>	SW20-500	LCN Installation	3025
<i>LCN System Checkout</i>	SW20-510	LCN Installation	3025
<i>Hewlett-Packard CD-ROM Documentation</i>			
<i>Hewlett-Packard Documentation at http://www.docs.hp.com</i>			

1. Universal Station^X Documentation

1.1 Overview

The documentation for the Universal Station^X Release 300 product is comprised of manuals published by Honeywell and Hewlett Packard. The HP documentation is available on a CD or at their website. The CD can be viewed using a CD-ROM reader installed on a U^XS, another Hewlett Packard workstation, or a PC with a CD-ROM drive. The documentation can be viewed or printed directly from the CD. Instructions for installation are provided with the Hewlett Packard CD.

1.2 Honeywell Documentation

Relevant Honeywell documentation for this release is listed in the Reference section of this manual. This manual is intended to supplement rather than duplicate the Hewlett-Packard documentation. This manual is not intended to be a standalone system administration manual—it should be used in conjunction with the HP documentation. This manual provides information specific to the U^XS, as well as information about basic HP-UX system administration tasks to help you get started.

1.3 Hewlett-Packard Documentation

HP web site

Hewlett-Packard HP-UX documentation can be accessed at <http://www.docs.hp.com>. This site contains a wealth of information including white papers and Frequently Asked Questions (FAQs).

HP-UX manual list by function

The numerous UNIX manuals can be grouped into functional categories to guide readers. Your manual requirement may vary depending on your job function, current level of UNIX expertise and level of interest.

Publication Title	System Administrators	General Users
<i>A Beginner's Guide to HP-UX</i>		√
<i>Administering ARPA Services</i>	√	
<i>Finding HP-UX Information Series 300/400/700</i>		√
<i>How HP-UX Works: Concepts for System Administrators</i>	√	
<i>HP-UX Reference (3 vols)</i>	√	
<i>HP-UX System Admin Tasks S700</i>	√	
<i>HP-UX System Security</i>	√	
<i>Inst/Update HP-UX S700</i>	√	
<i>Installing and Administering NFS</i>	√	
<i>Installing and Administering NS</i>	√	
<i>Installing Peripherals S700</i>	√	
<i>Master Index Series 700</i>	√	√
<i>Motif 1.1 Information Manual</i>	√	
<i>Networking Overview</i>	√	√
<i>Read Me Before Installing or Updating HP-UX</i>	√	
<i>Readme HP-UX 10.20 Series 700</i>	√	
<i>Remote Access: User's Guide</i>	√	
<i>Shells: User's Guide</i>	√	
<i>Solving HP-UX Problems</i>	√	
<i>Support Tools Mgr User's Guide</i>	√	

Publication Title	System Administrators	General Users
<i>Terminal Control: User's Guide</i>	√	
<i>Text Formatting: User's Guide</i>	√	
<i>The Ultimate Guide to the vi and ex Text Editors</i>	√	√
<i>Using ARPA Services</i>	√	
<i>Using HP-UX</i>	√	√
<i>Using Network Services</i>	√	
<i>Using NFS Services</i>	√	
<i>Using Serial Line IP Protocols</i>	√	
<i>Using the X Window System</i>	√	√

2. Universal Station^X Concepts

2.1 What is Universal Station^X

Long term plan

In today's competitive environment, maintaining control of costs and quality is paramount. Many process industry companies are formulating long-range plans to meet these objectives. The broad direction is to achieve low cost operations through the integration of process control and information system technologies. Honeywell's TDC (Total Distributed Control) system will be the foundation of the future CIM (Computer Integrated Manufacturing) system. This Honeywell system is called **TotalPlant**.

To further the evolution of this system, the Release 200 platform contained numerous upgrades to the Release 100 platform and added new features. These improvements were achieved without altering the basic elements of the Universal Station^X platform. This consistency of design preserved the investment in training, hardware, and software, proving "Consistent Evolution" to be a vital element in a cost-effective solution.

Release 300 of this product adds additional functionality including the upgrading of the operating system to HP-UX 10.20, which is fully year 2000 compliant.

Definition of Universal Station^X

An embedded open system environment provides support for display/import of data generated on non-LCN platforms. This TDC 3000^X /X Windows integrated platform enables software connections between the PIN (Plant Information Network) and the LCN (Local Control Network). Universal Station^X is the linking of open systems, X-Windows, and Motif to produce a visually integrated display of TotalPlant information.

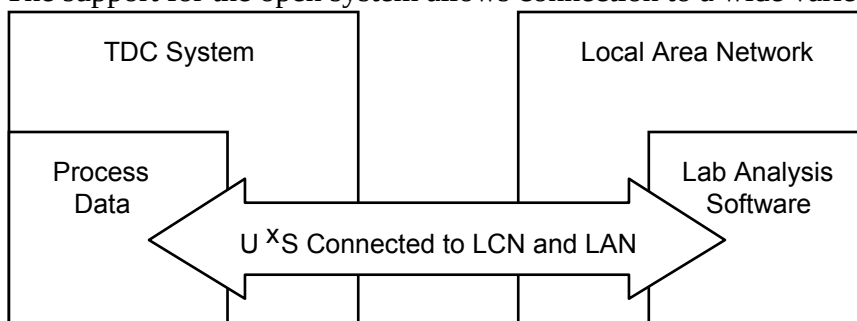
Universal Station^X features and benefits

Release 100 of the Universal Station^X product provided visual integration of plant information. Release 200 enabled U^XS functions to be added/alterd independent of a major LCN software release. Release 300 upgrades the operating system to HP-UX 10.20 making it fully Y2K compliant.

Features	Benefits
Native LCN Window	Menu bar for zoom and LCN window behavior options.
Key Repeat Timer	The keyboard has a repeat timer that can be configured
XSAM	Internationalization of text to display in the local (native) language.
Zooming Touch	Configuration for zooming touch in the LCN window .
Operator and parallel operator keyboard	The X server can be configured to always send key codes to the native LCN window.
X Server supports X11R5	Display of fonts that can be scaled, consistent color independent monitor type and support for internationalizing text to display in the local (native) language.
X Server extensions	Enhancements to X server functionality without affecting standard client operation.
Touchscreen Drag and Drop.	Emulates three button mouse, enabling text editing by drag and drop method. Also simplifies point and click.
Multiple PIN windows	Simultaneous access to multiple non-TDC computer systems
Fail Safe Fall Back	Ensures the “ <i>view to the valve.</i> ”
System Security	Limits access to authorized users. TDC system isolated from open system.

Universal Station^X example

The support for the open system allows connection to a wide variety of hardware platforms.



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Currently, information on the LAN (Local Area Network) can be displayed or updated at the Universal Station^X. This allows operators to view lab analysis data on the same screen as the process data. By integrating this data, the operator has fewer distractions and requires less training (only one computer system). A typical control room operator may have two or three computers to monitor (each holding some essential element of process data). When the operator's attention is diverted from the primary TDC display the potential for problems increases.

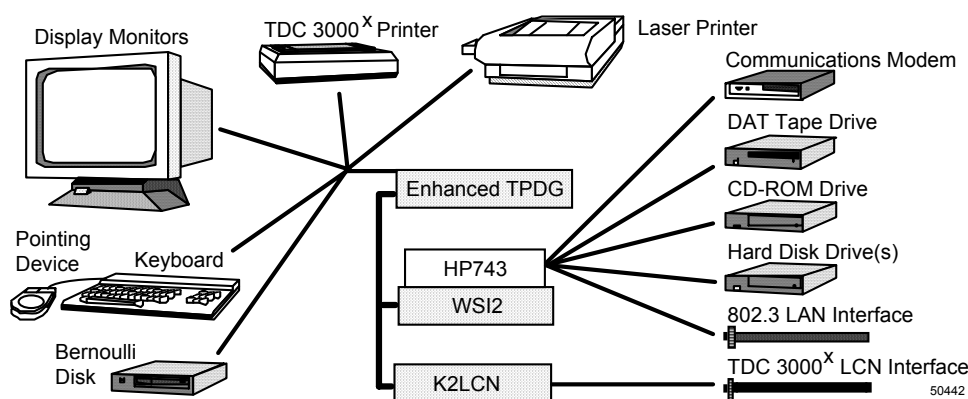
2.2 Universal Station^X Hardware

Components

The Universal Station^X consists of a tightly integrated TDC 3000^X Universal Station and an open system processor environment in a coprocessor architecture. The Universal Station^X contains the following components:

- Standard US furniture cabinet
- Standard 5-slot module
- Electronics package
- High resolution monitor (touchscreen optional)
- Hard drive connected to coprocessor
- DAT (Digital Audio Tape) drive connected to coprocessor (one DAT required per system).

Component diagram



Electronics package

The electronics package consists of the following:

- Standard K2LCN-8 board
- Workstation Interface board (WSI board)
- Turbo Peripheral Display Generator (TPDG board)
- I/O boards for each of the boards above
- 8 ampere power supply

Coprocessor media

The hard drive and a DAT (Digital Audio Tape) drive contain these Universal Station^X files:

- Open system filesystem
- X Window files
- TPDG download code
- Universal Station^X (UNPX) personality download code

This media is connected to the Open system coprocessor and is not accessible by the LCN filesystem.

Specific details

For more specific hardware information, refer to the *Universal Station^X Service* manual.

3. Window Operations

3.1 Window Manager

X window system

The X Window system provides standards by which computers of dissimilar types can exchange window displays. This allows computer systems to execute an application and send the display of that application to other computer systems that support the X window graphical standard. This standard integrated the concept of a window manager. The window manager allows manipulation of the X window displays.

Consistent look

The Universal Station^X utilizes Motif Window Manager to provide a consistent look to all applications displayed on the Universal Station^X. This software provides functions for opening, closing, moving, resizing, and indicating the active window. If you have used Microsoft Windows you will immediately recognize the similarities.

The Window Manager attributes are preset, and cannot be altered by the user. System administrators may provide custom configuration of window attributes on a “system wide” or “user by user” basis.

Functions provided

The Motif Window Manager applies the window interface attributes independent of the application running within a window. The window interface affects only the manner in which that window is displayed. It will provide such elements as title bars, scroll bars, resizing, control of overlapping windows, and the ability to represent a window as an icon. These elements allow manipulation of multiple windows, independent of their application or the underlying computer platform/operating system executing the application. This allows one window interface to control the display of all windows appearing on the Universal Station^X screen.

3.2 Cursor Usage

Cursor types

The Window Manager uses various cursor types to depict the type of operation being performed. The types are:

- pointing  or  or 
- working 
- moving 
- resizing  or  or  or  or  or  or  or 

3.3 Altering Window Size

Changing window size

Where you grab the window's frame determines how the window will be resized. The dimensions (width x height) of the window being resized is displayed in the center of the screen. The measurement is in characters (across) and lines (down) for terminal windows; other types are measured in pixels (dots on the screen).

- ⇒ **Note:** When resizing the LCN Native Window, it is necessary to click the bottom right corner of the window border to cause the LCN display to resize. The LCN processing continues without problem, but the display may be only partially visible.

Use the following to change the window size:

If you want to resize the window . . .	Point to, depress, and hold the mouse button and drag. . .
vertically from the . . .	
top	top of the frame, above the title bar
bottom	bottom of the frame
horizontally from the . . .	
right	right side of the frame
left	left side of the frame
diagonally from the . . .	
bottom left corner	frame's lower left corner
top left	frame's upper left corner
top right	frame's upper right corner
bottom right	frame's lower right corner

Zoom enhancements

Release 200 provided several enhancements to the Native LCN Window. A menu bar with “size” and “options” menus was added to the Native LCN Window’s window border. It is important to remember the LCN generates a constant display size that is resized to fit the window border by the TPDG board. Resize is not an LCN function, but performing a resize could (depending on circumstances) trigger other changes to the Native LCN window appearance.

The “size” menu is used to select the desired size of the Native LCN window and has the following choices:

- **Small** — standard one quarter screen size
- **Large** — full screen size

Native LCN window Option Menu

The “Options” menu controls the behavior of the Native LCN window and has the following items:

- **Allow Partial Window Overlay** — configures the ability to partially obscure the Native LCN window with other windows.
- **Allow Total Window Overlay** — configures the ability to totally obscure (overlay) the Native LCN window with other windows. You currently must select the window a second time for it to overlay the Native LCN window.
- **Zoom Touch active in Full Screen mode** — Enables Zoom Touch when Native LCN Window is displayed at full screen.
- **Automatic Resize** — maximizes Native LCN window and sets display to full screen.

3.4 Minimizing/Maximizing Windows

Minimize a window

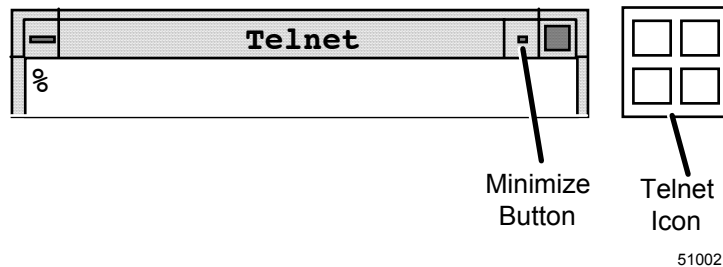
To reduce workspace clutter, you may change a window to an icon. An icon is a graphical representation of the application executing in that window. This reduction to an icon is called “minimizing” a window. Applications executing in a minimized window continue to execute until the task is completed or until they require user input.

There are several methods to minimize a window:

- Use the **Minimize** button on the windows Title Bar
- Use the **Minimize** selection from the Window Menu

A window that has been minimized may be returned to the window form using several methods.

- Double-Click (click twice in rapid succession) the icon.
- Use the **Maximize** selection from the minimized window icon menu.



Maximize a window

Clicking on the Maximize button will enlarge the window to cover the entire monitor. Clicking on the button a second time will return the window size to the original dimensions.

⇒ **Note:** When resizing the LCN Native Window, it is necessary to click on the bottom right corner of the window border to cause the LCN display to resize. The LCN processing continues without problem, but the display may be only partially visible.

3.5 Using the Window Menu

Window menu

Each window has a window menu containing functions for controlling the window. To access the window menu, point to the menu target and then press and hold the select button. The menu is displayed as long as the select button is held down. To select a menu item, drag the cursor down the list and release the select button when the appropriate item is highlighted.



Menu
Button

Window
Menu

Restore	Alt + F5
Move	Alt + F7
Size	Alt + F8
Minimize	Alt + F9
Maximize	Alt + F10
Lower	Alt + F3
Close	Alt + F4

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Window Operations – Using the Window Menu

To perform the following, select the appropriate menu item:

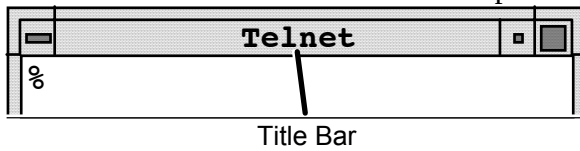
To do this . . .	Select . . .
Restore a window from an icon or after maximizing.	Restore
Change the location of a window.	Move
Change the size of a window.	Size
Shrink the window to its icon representation.	Minimize
Enlarge the window to cover the entire root window.	Maximize
Send a window to the back or bottom of the window stack.	Lower
Immediately close the window and make it disappear.	Close

3.6 Repositioning Windows

Moving windows

Windows can be positioned anywhere on the screen, with one exception. The Native LCN window is initially configured (for all users except engineer group members) to inhibit its ability to be partially obscured or overlapped by other windows. This attribute can be configured for users of all access levels and is covered in subsection 6.3, *Configure Native Window Behavior*.

To move a window, click and drag on the window's title bar. While moving, an outline of the window is shown for positioning purposes. The window will move to this new location if the select button is released at the current position.



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3.7 Workspace Menu Access

Workspace menu

To reduce the user interaction with the open system environment, a workspace menu (sometimes called the root menu) is included. This menu is accessed by holding down the select button within the workspace (outside all windows). You can select a menu item by dragging the pointer down the menu list and releasing the select button when that item is highlighted.

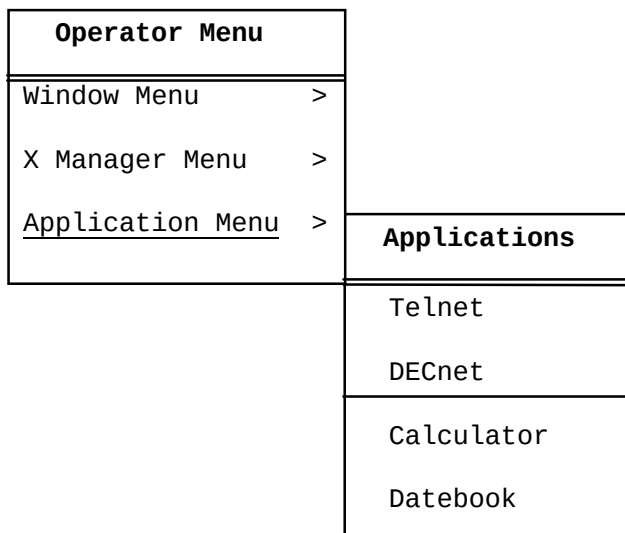
Operator Menu	Engineer Menu
Window Menu >	Window Menu >
X Manager Menu >	X Manager Menu >
Application Menu >	Application Menu >
	System Menu >

Supervisor Menu	View Only Menu
Window Menu >	Window Menu >
X Manager Menu >	X Manager Menu >
Application Menu >	

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Hierarchical menus

The menu uses hierarchical (nested) menu items indicated by an arrow to the right of the item that has another level of menu. To access these nested menus hold the select button and scroll to the appropriate menu; the nested menu item will appear. Moving the cursor onto this new menu will allow selections to be made from this newly visible list of items.



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Access levels

The menu is preconfigured according to the access level, which is assigned by the system administrator. When a user is added to the coprocessor, their user-login name is associated to a group of users. This association provides security restrictions by displaying menus specific to that group's access authorization.

The access levels are

- operator
- supervisor
- engineer
- view Only

3.8 Operator Workspace Menu

Operator Menu items

- The Operator Menu contains the following:
- Window Menu - a nested menu
- X Manager Menu - a nested menu

- `Application Menu` - a nested menu

Operator Menu	
Window Menu	>
X Manager Menu	>
Application Menu	>

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Window Menu items

The operator's `Window Menu` contains the following:

- `Shuffle` – Shuffles the position (*Up/Down*) of the active window.
- `Other Screen` – This menu item toggles between dual screens allowing one pointing device and one set of keyboards to control both screens.
- `Keyboard Focus Policy` – Configures the X Windows attribute (`Explicit/Pointer`) to identify the active window. Either a mouse click (`Explicit`) or cursor within the window (`Pointer`) is required to identify the active window.
- `Window Print` – Prints the contents of the selected window.

X Manager items

- The operator's `x Manager` Menu contains the following:
- `X Host Authorization` - Authorizes remote computing resources to display X applications on this station. Configured for each individual user.
- `Change User Password` - allows the user to change their login password
- `Refresh` - Performs a redraw of the X application windows.
- `Restart` - Restarts the Motif Window Manager; used to force a reset of Motif Window Manager.
- `Logout` - Sends a kill message to all X hosts to close open applications, logs the user out, and idles the coprocessor.

Application Menu

- The operator's `Application` Menu contains the following:
- `Telnet` - accesses the TELNET TCP/IP protocol
- `DECnet` - accesses the DECnet protocol
- `Calculator` - a full function calculator application
- `Datebook` - a calendar/datebook application

3.9 Supervisor Workspace Menu

Supervisor Menu items

The Supervisor Menu contains the following:

- Window Menu - a nested menu
- X Manager Menu – a nested menu
- Application Menu – a nested menu

Supervisor Menu	
Window Menu	>
X Manager Menu	>
Application Menu	>

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Window Menu items

The supervisor's Window Menu contains the following:

- Shuffle - Shuffles the position (*Up/Down*) of the active window.
- Other Screen - This menu item toggles between dual screens allowing one pointing device and one set of keyboards to control both screens.
- Keyboard Focus Policy - Configures the X Windows attribute (*Explicit/Pointer*) to identify the active window. Either a mouse click (*Explicit*) or cursor within the window (*Pointer*) is required to identify the active window.
- Window Print - Prints the contents of the selected window.

X Manager items

- The supervisor's **X Manager Menu** contains the following:
- **X Host Authorization** - Authorizes remote computing resources to display X applications on this station. Configured for each individual user.
- **Change User Password** - allows the user to change their login password
- **Refresh** - Performs a redraw of the X application windows.
- **Restart** - Restarts the Motif Window Manager; used to force a reset of Motif Window Manager.
- **Logout** - Sends a kill message to all X hosts to close open applications, logs the user out, and idles the coprocessor.

Application Menu

The supervisor's **Application Menu** contains the following:

- **Telnet** - accesses the TELNET TCP/IP protocol
- **DECnet** - accesses the DECnet protocol
- **Calculator** - a full function calculator application
- **Datebook** - a calendar/datebook application

3.10 Engineer Workspace Menu

Engineer Menu items

The Engineer Menu contains the following:

- Window Menu - a nested menu
- X Menu - a nested menu
- Application - a nested menu
- System Menu - a nested menu

Engineer Menu	
Window Menu	>
X Manager Menu	>
Application Menu	>
System Menu	>

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Window Menu items

The engineer's Window Menu contains the following:

- Shuffle - Shuffles the position (*Up/Down*) of the active window.
- Other Screen - This menu item toggles between dual screens allowing one pointing device and one set of keyboards to control both screens.
- Keyboard Focus Policy - Configures the X Windows attribute (*Explicit/Pointer*) to identify the active window. Either a mouse click (*Explicit*) or cursor within the window (*Pointer*) is required to identify the active window.
- Window Print - Prints the contents of the selected window.

X Manager Menu items

The engineer's x Manager Menu contains the following:

- **X Host Authorization** - Authorizes remote computing resources to display X applications on this station. Configured for each individual user.
- **Change User Password** - allows the engineer to change a user's login password
- **Refresh** - Performs a redraw of the X application windows.
- **Restart** - Restarts the Motif Window Manager; used to force a reset of Motif Window Manager.
- **Logout** - Sends a kill message to all X hosts to close open applications, logs the user out, and idles the coprocessor.

Application Menu items

The engineer's Application Menu contains the following:

- **Telnet** - accesses the TELNET TCP/IP protocol
- **DECnet** - accesses the DECnet protocol
- **Calculator** - a full function calculator application
- **Datebook** - a calendar/datebook application
- **UNIX Load** - a bar chart display of UNIX processor loading

System Menu items

The engineer's System Commands menu contains the following:

- **Configuration** - a nested menu of configuration items
- **Update Software** - used to update coprocessor software
- **Add/Remove Users** - a coprocessor user registration application
- **Backup/Restore** - used to backup/restore the coprocessor software

Configuration Menu items

The engineer's Configuration menu contains the following:

- `IWSM Configuration` - Shutdown WSI and Native Mode access
- `System Configuration` - used to configure the coprocessor LAN environment
- `System Terminal` - used to open a terminal session window.
- `Set Time/Timezone` - sets the coprocessor time, date, and timezone

3.11 View-Only Workspace Menu

X Menu items

The View-Only X Menu contains the following:

- Window Menu - a nested menu
- X Manager Menu - a nested menu

View-Only Menu	
Window Menu	>
X Manager Menu	>

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Window Menu items

The Window Menu contains the following:

- Shuffle - Shuffles the position (*Up/Down*) of the active window.
- Other Screen - This menu item toggles between dual screens allowing one pointing device and one set of keyboards to control both screens.
- Keyboard Focus Policy - Configures the X Windows attribute (*Explicit/Pointer*) to identify the active window. Either a mouse click (*Explicit*) or cursor within the window (*Pointer*) is required to identify the active window.
- Window Print - Prints the contents of the selected window.

X Manager Menu items

The X Manager Menu contains the following:

- **X Host Authorization** - Authorizes remote computing resources to display X applications on this station. Configured for each individual user.
- **Change User Password** - Allows the user to change their login password.
- **Refresh** - Performs a redraw of the X application windows.
- **Restart** - Restarts the Motif Window Manager; used to force a reset of Motif Window Manager.
- **Logout** - Sends a kill message to all X hosts to close open applications, logs the user out, and idles the coprocessor.

3.12 Using the Touchscreen

Display resolution

The high-resolution display required a change to touchscreen operation. Software is provided to zoom the screen display at 2 times original size. This zoom touchscreen aids in the selection of small targets.

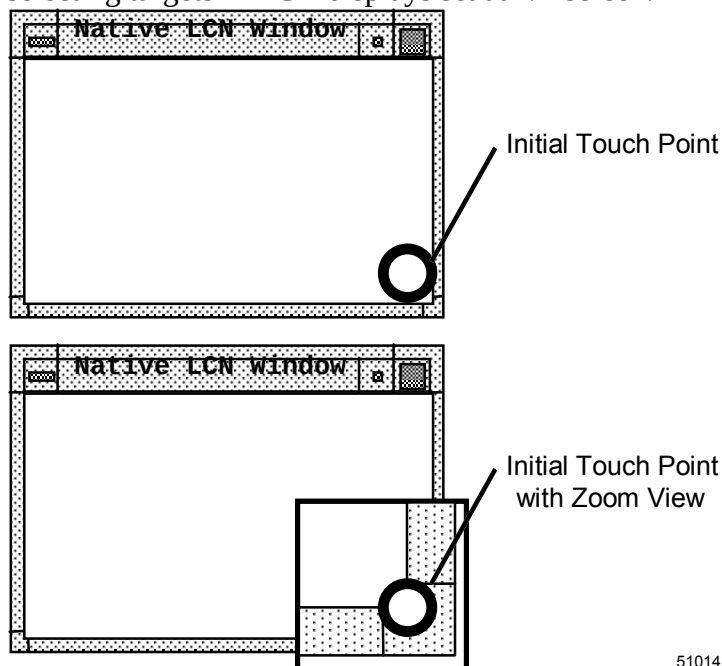
Zoom feature availability

The zoom feature is only available when a user is logged into the coprocessor environment. The software action is performed as part of the X window environment and does not effect operation of the application within the window.

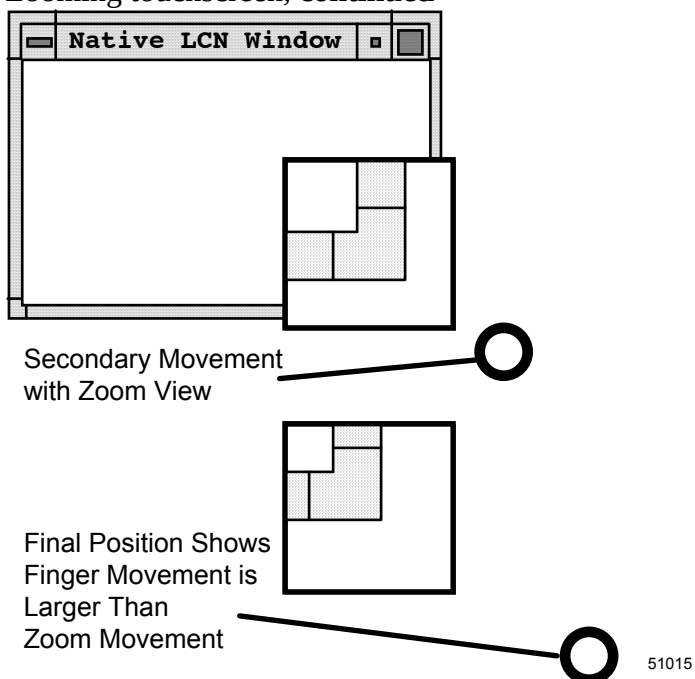
Zooming touchscreen

The touchscreen on the Universal Station^X is supplemented by software to aid in selecting small targets in a high resolution display.

When the screen is touched, a zoom window boundary is created approximately 3 inches around the initial touch point. A zoomed-in view (2x zoom) doubles the actual size of the display inside the zoom window boundary. Moving the finger 2 inches to the right moves the image inside the zoom window only 1 inch. This provides the necessary resolution for selecting targets in LCN displays set at 1/4 screen.



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Zooming touchscreen, **continued**

The proper method of using the zoom touchscreen is to place your finger on (or as close to) the desired target as possible. This is to reduce the amount of finger movement necessary allowing selection of targets close to the screen border. It is important to watch the zoomed view, as the 2:1 ratio of finger to image movement is initially confusing.

3.13 Configuring and Using Dual Screens

Overview

The Dual CRT function of the U^XS release 200 software allows for two CRTs to be controlled by one set of U^XS electronics. When the dual screen feature is selected for the U^XS software, it can be configured to be in an over-under configuration or in a side-by-side configuration. When in the over-under configuration, screen 0 is located on the bottom with screen 1 being on top. When in the side-by-side configuration, screen 0 is located on the left and screen 1 is located on the right when facing the CRTs.

Input devices

When using the Dual CRT function of U^XS release 200, a single mouse/trackball and set of keyboards are used for interaction with the system on either screen. A Motif Window Manager (mwm) menu selection can also be configured to “hop” the cursor between screens.

Touchscreen

The Dual CRT function of U^XS release 200 allows for a touchscreen to be connected to screen 0. There is no hardware or software support to allow for connection of a touchscreen to screen 1.

Limitations

When using the Dual-CRT function of U^XS release 200, the native LCN window can only be displayed on screen 0. All other applications may be displayed on either screen 0 or screen 1.

Configuration file setup

Three files are included in the /usr/lib/X11 directory for the configuration of Dual CRT functionality. They are:

- *usxserverrc.OnlyOneScreen* (for single crt stations)
- *usxserverrc.OverAndUnder* (for vertically oriented screens)
- *usxserverrc.SideBySide* (for horizontally oriented screens)

One of these files **must** be used and linked to the file name “*usxserverrc*” as part of the system startup configuration.

Example:

```
ln -s usxserverrc.OverAndUnder usxserverrc
```

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is required.

WARNING

Take care when editing this file. If you make an error when editing file, it may stop the X-server from running (when you log out). In this case, the login banner will not be displayed. Without the login banner, you will not be able to login to the Universal Station^X to correct your mistake. To correct your mistake, you will have to login from a remote device, or connect a console terminal to the WSI2 I/O board.

Link usxserverrc to the appropriate screen configuration file

Perform the following steps to configure the system (X server) for the appropriate screen configuration:

Step	Action
1	Select System Menu->Configuration->System Terminal from the Workspace Menu.
2	Enter the coprocessor (root) password and press <RETURN>. Note: Only <RETURN> is necessary on a new station.
3	From the shell prompt, enter <code>cd /usr/lib/X11</code> and press <RETURN>.
4	Unlink <code>usxserverrc</code> from the file to which it is currently linked by entering <code>unlink usxserverrc</code> and pressing <RETURN>.
5	From the shell prompt, use the link command to link <code>usxserverrc</code> to the appropriate screen configuration file. Use one of the three commands that follow: <code>ln -s usxserverrc.OnlyOneScreen usxserverrc</code> <code>ln -s usxserverrc.OverAndUnder usxserverrc</code> <code>ln -s usxserverrc.SideBySide usxserverrc</code>

Setup Motif for multiple screens

Perform the following steps to configure the Motif window manager for multiple screens.

Step	Action
1	Edit <code>/usr/lib/X11/system.Xdefaults</code> to change (if necessary) the following resource to be setup as true: <code>Mwm*multiScreen: true</code> The following commands can be used to edit the file: Note: Unless specified, do NOT press <RETURN> a) <code>vi system.Xdefaults</code> and press <RETURN> b) <code>/multi</code> and press <RETURN> c) <code>W</code> d) <code>cw</code> e) <code>true</code> f) <ESC> g) <code>:wq!</code> and press <RETURN> Note: You must logout and login to make your screen configuration effective.
2	Enter <code>exit</code> and press <RETURN>.

Command line use

To display a window on the second screen you must use the “-display” option when invoking the application. For example:

```
xclock -display unix:0.1 <RETURN>
```

This will start the application “xclock” on the second screen (screen 1). If you use “-display unix:0.0”, the application will be displayed on the first screen (screen 0).

Refer the Hewlett-Packard manual “*Using the X Window System*” for more detailed information on X Window usage.

.mwmrc file use

You can add “-display unix:0.1” to the f.exec definition for the Motif menu items.

Default as screen 0:

```
"Calculator"f.exec "xcalc -geom 220x280-160+4 -fn 9x15 &"
```

With -display added to go to screen 1:

```
"Calculator"f.exec "xcalc -display unix:0.1 -geom 220x280-160+4  
-fn 9x15 &"
```

⇒ **Note:** If an application is started from a Motif menu and there is no definition for the display to use, the application will start up on the screen from which it was invoked.

4. Universal Station^x Operations

4.1 Power-On

Apply power

Apply power using the following steps:

Step	Action
1	Place the station circuit breaker in the ON position.
2	Open the rear access door to gain access to the 5-slot module.
3	Apply power to the station using the power switch on the 5-slot module faceplate.
4	Observe the K2LCN board LEDs during the power on self-test. If no errors are detected, the appropriate LCN node address is displayed.
5	<u>If the appropriate LCN node address is displayed</u> Proceed with Step 6. <u>If another number is displayed</u> Record the error number for maintenance personnel to assist in correcting the fault. <u>If the banner does appear within 5 minutes</u> Proceed with Step 8 to shut down the coprocessor. <u>If the Login banner has not appeared</u> Proceed with Step 11.
6	The Login banner should appear within 5 minutes.
7	<u>If the Login banner does appear</u> Proceed to subsection 4.2, <i>Accessing Coprocessor Functions</i> . <u>If the Login banner has not appeared</u> Proceed with Step 11.
8	Position the cursor in the Login box and type <i>username</i> <RETURN> Note: The user must be authorized to shut down the WSI.
9	Position the cursor in the Password box and type <i>password</i> <RETURN> Note: Replace <i>password</i> with your password.

4.2 Accessing Universal Station^X Coprocessor Functions

Coprocessor access banner

After power ON (3 to 5 minutes) the Universal Station^X will display a banner on the lower portion of the screen.

This banner allows access to:

- X windows environment
- Shutdown of the coprocessor
- Universal Station^X Native Mode operation.

These items are user and password restricted to maintain security and restrict access to the coprocessor or LAN environment.

Using the access banner

Using the touchscreen:

Step	Action
1	Touch the <code>Login</code> field.
2	Type your <i>username</i> <RETURN> .
3	Type your <i>password</i> .
4	<u>To Login:</u> Press <RETURN> <u>To Shutdown the Coprocessor</u> Touch the <code>Shutdown WSI</code> target and select it with the space bar. <u>To revert to Standard Universal Station Operation</u> Touch the <code>Native Mode</code> target and select it with the space bar.

Using the mouse or trackball:

Step	Action
1	Select the Login field by placing the cursor over the field.
2	Type your <i>username</i> <RETURN> .
3	Type your <i>password</i> .
4	<u>To Login</u> Press <RETURN> <u>To Shutdown the Coprocessor</u> Select the Shutdown WSI target. <u>To revert to Standard Universal Station Operation</u> Select the Native Mode target.

4.3 Loading a Personality to the LCN Processor

Capabilities

The Universal Personality has been encapsulated in a “Windows environment” to form the UNPX Personality stored on the coprocessor hard drive. This integration enables the Universal Station^X to function as either a Universal Station^X or a standard TDC 3000^X Universal Station.

The X window capabilities are available only when functioning as a Universal Station^X. As a Universal Station^X, the presentation of all displays will be controlled by the X windows/Motif Window Manager to provide the addition of such items as title bar, zoom box, and scroll bars.

Hardware tests

A fully operational Universal Station^X system can use the UNPX personality. When power is applied, the system performs a series of internal self-tests. These tests check the presence and condition of both the TDC and coprocessor hardware. The results determine the loading action available when <Load> is pressed.

If test result is ...	then loading action available is ...
TDC and coprocessor hardware OK	UNPX personality from hard drive or UNP, ENG, or OPR from History Module
TDC hardware OK/coprocessor test failure	UNP, ENG, or OPR from History Module or Removable Media
TDC and coprocessor hardware failure	No loading performed

Loading the personality

The Universal Station^X may be loaded from several sources:

- History Module
- Removable Media
- Internal coprocessor hard drive

The Universal Station^X can be loaded from other stations (Universal Station^X or US) on the LCN. However, when loaded locally, the hardware tests will control the appropriate prompt. For example, failure of the coprocessor will inhibit the load prompt from displaying the appropriate choice (W).

The four personalities available for loading on the Universal Station^X are:

- OPR
- ENG
- UNP
- UNPX

The first three are available from the History Module or removable media, the last (UNPX) can be loaded only from the coprocessor. To get full Universal Station^X functionality, the LCN processor must be loaded with the UNPX personality. This allows data (point.parameter) to be transferred from the “LCN side” to the “UNIX side”. Logging into a windowing environment is available regardless of which LCN personality is loaded.

ATTENTION

Touch screen functionality is not available during the load of the LCN processor.

Upgrading from K2LCN to K4LCN

If you are upgrading from a K2LCN to a K4LCN, you need to perform the following commands in order for the loading of the personality from “W” to complete successfully. These commands should be done prior to shutting down the system for the K2LCN to K4LCN board swap, or if this has already been down, prior to pressing the LOAD key. These commands can only be executed by “root”.

Step	Action
1	Make sure you are in the appropriate directory. <code>cd /etc/opt/TDC_Open/common</code>
2	Remove the old personality file. <code>rm personality.config</code>
3	Establish a symbolic link from the new personality file(personality.config.usxk) to the active personality file(personality.config). The format for the command is: <code>ln -s filename linkname</code> Example: <code>ln -s /opt/TDC_Open/common/newconfig/personality.config.usxk personality.config</code>

Load the LCN processor

Use the following procedure to load the LCN processor. The touch screen is not active during this process and should not be used.

Step	Action
1	<u>If the access banner is on the screen</u> Press <Load> on the operator keyboard. <u>If you are already logged in</u> Select the Native LCN Window, press <Load> on the operator keyboard.
2	At the prompt W, N, 1, 2, 3, 4, X type W,Press <Enter>
3	The load modules will be displayed as the LCN processor is loaded.
4	At the prompt NCF? N,1,2,3,4,X? type N, Press <Enter>
5	The station will complete the load process in 1 to 2 minutes.

4.4 Printing from the Universal Station^X

Two printers

The Universal Station^X has two printer connections.

The standard LCN printer connection provided to print LCN displays and reports is configured as a standard Universal Station printer. This printer is functional whenever the station is in operation. The LCN printer can be connected directly to a Universal Station^X or assigned from another station in the console (US or Universal Station^X). Standard Universal Station^X operating procedures apply to the printing of reports, alarm summaries, configuration data, print displays, etc.

The coprocessor has a printer connection provided to print window dumps of X Window applications. This printer is available only when the user is logged into the coprocessor; however, printing will still continue after logging out. This printer will not print the various Universal Station Printer functions of alarm summaries, reports, etc. The coprocessor printer can be directly attached to this Universal Station^X or shared across the LAN from another Universal Station^X.

Printing the LCN window

The LCN processor must be active to print. The user can be logged into the coprocessor, but it is not required to print LCN data.

To print the LCN window on the LCN printer, use the following:

Step	Action
1	<u>If logged into the coprocessor</u> Select the LCN window.
2	Press the Print Display key.

Printing X Windows

When the user is logged into the coprocessor, the coprocessor printer can be used to print any window contents. To print to a shared printer, the coprocessor of the Universal Station^X sharing the printer must be active (login banner displayed or a user logged in).

To print any X Window on the coprocessor printer, use the following:

Step	Action
1	Select window <code>Print</code> from the workspace menu. Note: The cursor will change to a cross-hair pointer.
2	Position the cursor in the appropriate window to print.
3	Click the select button.

4.5 Failsafe Operation

Failsafe fall back

The Universal Station^X is designed to be failsafe. Should an error occur in the coprocessor, the Universal Station^X reverts to a standard TDC Universal Station^X hardware/software configuration. If this should occur, the LCN display will revert to full screen and the following message will be displayed in place of the Login banner:

Coprocessor has failed to start, Call maintenance

When functioning as a standard Universal Station, you will be unable to access LAN (Local Area Network) resident applications.

Native mode

The Universal Station^X can be configured as a standard Universal Station by selecting Native Mode from the Login banner. To enter Native Mode, the user must be authorized using the procedures covered in *IWSM Configuration, Universal Station^X System Administration Manual*.

Manual shutdown

The coprocessor can be shut down after power-on using the Shutdown WSI target located on the login prompt. This action can also be performed by logging out of the open system session and waiting for the login prompt to appear.

To invoke the failsafe condition:

- At the login prompt, enter your login name/password and select the Shutdown WSI target.

Shutdown restriction

To shutdown the WSI, the user must be authorized using the procedures covered in “IWSM Configuration,” *Universal Station^X System Administration Manual*.

Restarting after shutdown

After the coprocessor has been shutdown, a status message will appear in the Login banner area. The message is:

The coprocessor is idle, select the box to reset.

There is a red box to the right of the status message. Using the mouse, click on the box to reset the coprocessor.

4.6 Universal Station^X Display Changes

TDC display changes

Several changes are apparent when operating from the Universal Station^X. Some changes are due to TDC software Release 430, and are not Universal Station^X inherent.

- **Console and Node Status Displays**
A field for the personality type was added. The node status is not overwritten by personality type when status is OK.
- **Console Status Display**
Ability to choose the personality which is loaded to a Universal Station^X was added. The `autoload_net` target determines the default personality to load by testing the hardware.
- **Real Time Journal Changes**
The System Maintenance Journal is updated with additional messages to support the new TPDG hardware. The System Status Change Journal will be updated with a new message—State Change Notification (SCN) to establish the personality subtype in a node.

4.7 Window Operation

Background

The Universal Station^X is configured to launch the X window environment upon login. Once the system has initialized, the primary LCN display window appears. At this time you can start an X window session with LAN resident computing resources.

Prerequisites

The LAN computing resources must be configured to allow access from the Universal Station^X. In general terms, the following information is required by the LAN hardware and/or application software:

- Universal Station^X internet address
- System name
- User name/password for its login

Choosing emulators and protocols

To access LAN-resident applications you will use a communication protocol (either TCP/IP or DECnet), and a terminal emulator (either xterm or vt340). The DECnet protocol and the vt340 terminal emulator are optional packages on the Universal Station^X.

Terminal Emulator	Protocol	Menu Selection	Command(s)
vt102 (xterm)	Telnet TCP/IP	Telnet	open <i>hostname</i>
vt102 (xterm)	DECnet	Telnet	! sethost <i>hostname</i>
vt340	Telnet TCP/IP	DECnet	telnet <i>hostname</i>
vt340	DECnet	DECnet	sethost <i>hostname</i>

Accessing the LAN

LAN resident applications may not be Honeywell supplied; therefore, these access procedures are general in nature.

Step	Action
1	Login to the Universal Station ^X .
2	Determine the appropriate terminal emulator and protocol.
3	Make the appropriate menu selection (to select the desired terminal emulator).
4	Enter the appropriate command(s) (to initiate a login sequence to the remote LAN device using the desired protocol).
5	Log in to the remote LAN device.
6	Enter the appropriate command(s) to launch the application in the remote LAN device.

4.8 System Security

TDC isolated

The TDC operating system is isolated and inaccessible to the coprocessor. This eliminates the possibility of misuse/tampering with the process control system by unauthorized personnel who may gain access to the Local Area Network.

The full implementation of HP-UX version 10.20 security features are provided. System security is controlled through the configuration and login utilities.

Configuration

Configuration parameters ensure only designated computer systems will be able to login to the Universal Station^X. This allows multiple computer systems to exchange data transparently while ensuring data integrity through limited authorized access.

The coprocessor password should be changed and recorded during initial configuration. This password is the highest level of system access and system security must be enforced by good password discipline.

Login

A password controlled login utility protects against unauthorized Universal Station^X coprocessor usage. This utility configures access dependent on the user-login identification. The system customizes the workspace menu as appropriate to the access level of the user. Access to additional items can be provided to individual users by modifying the `.mwmrc` file in the users home directory.

Access keylock versus coprocessor login

This login utility performs a function similar to the Access Level Keylock on the standard Universal Station; however, the coprocessor login is entirely independent from the keylock mechanism. System security may be compromised if the user leaves the immediate area without logging out of the coprocessor environment.

Remove the access level key and logout of the coprocessor environment to prevent unauthorized access to the LCN system.

Emphasize security

Users should be reminded of the potential security risks. A periodic review of the system security, security procedures, and system users will help to raise awareness. Remind users that the password is their computer signature. Emphasize the importance of periodically changing this password.

Coprocessor access by group

The restrictions placed upon access levels are:

- Operator—able to configure their own environment, use telnet, and the optional VT340 DECnet communications
- Supervisor—able to configure their own environment, use telnet, and the optional VT340 DECnet communications
- Engineer—able to configure their own environment, use telnet, the optional VT340 S/W, coprocessor configuration (requires coprocessor password), and perform periodic backups of the coprocessor file system
- View Only—able to configure their own environment.

⇒ **Note:** User-configurable items are: X Hosts, allowPartiallyObscured Native Window, and Keyboard Focus Policy.

Remote user configuration

To configure the remote user coprocessor security features, review the procedures, gather the required information, and perform all tasks in Section 5, “*Configure Security for Remote Users*”.

4.9 Shut Down the Universal Station^X

Background

To prevent damage to the UNIX filesystem, the station must be shut down using a special sequence. The shutdown procedure must be followed before power is removed from the station. Only those users configured in IWSM configuration are allowed to shutdown the coprocessor.

Failure to perform the proper shutdown procedure could result in damage to the coprocessor filesystem. The next time the station is powered up, it may take up to 5 minutes to perform automatic repairs of the coprocessor filesystem.

WARNING

Use caution in cycling power on and off. If this is done while a repair is being made, the filesystem may be left in a non-repairable state. This would require reloading the filesystem backup from tape and possibly require reformatting or replacement of the hard drive.

Procedure

If you are currently logged into the Universal Station^X, use the following procedure:

Step	Action
1	Select Logout from the root menu.
2	At the prompt <code>Quit Mwm?</code> , select the OK target.
3	When login menu appears, enter your username and password. Note: Do not press <RETURN> after entering your password.
4	Select SHUTDOWN WSI target.
5	Press the <code>CONSOLE STATUS</code> key.
6	Select the node number of the appropriate Universal Station ^X .
7	Select the SHUTDOWN target.
8	Press ENTER to confirm the shutdown request.

Universal StationX Operations – Shut Down the Universal StationX

If no user is logged into the Universal Station^X, use the following procedure:

Step	Action
1	When login menu appears, enter your username and password. Note: Do not press <RETURN> after entering your password.
2	Select the SHUTDOWN WSI target from the login menu.
3	Press the CONSOLE STATUS key.
4	Select the node number of the appropriate Universal Station ^X .
5	Select the SHUTDOWN target.
6	Press ENTER to confirm the shutdown request.

4.10 Universal Station^X Restrictions

Shutdown procedure

Because the coprocessor operating system is incorporated into the Universal Station^X, the station must be turned off using a special sequence. The shutdown procedure must be followed any time power is to be removed from the station.

Improper shutdown procedure could damage the coprocessor filesystem. The next time the station is powered up it may take 5 minutes to perform automatic repairs to the coprocessor filesystem. **Use caution in cycling power on and off.** If this is done while a repair is being made, the filesystem may be left in a non-repairable state. This would require reloading the filesystem backup from tape, and possibly require reformatting or replacement of the hard drive.

Coprocessor red box target

Under most circumstances the coprocessor “red box” target should not be selected as the system is in the process of a self reset and the red box will go away on its own. Selecting the red box prematurely can cause UNIX related file problems. If you feel the red box target has been displayed for an unusually long time, wait a minimum of 10 total minutes before selecting it for a system reset.

4.11 Technical Assistance

If you need assistance

If you need technical assistance, contact your local Honeywell Service Organization, as explained in the following paragraphs.

International customers

Outside of the United States, contact your local Honeywell Service Organization. If you are not sure of the location or telephone number, call your Honeywell representative for information.

Customers inside the United States

Within the United States, call the Solutions Support Center (SCC) at the toll free number 1-800-822-7673.

Arizona customers

Within Arizona, the local number for SCC is 602-863-5558.

Services provided

Calls to SCC are answered by a dispatcher from 7:00 A.M. to 5:00 P.M., Mountain Standard Time (6:00 A.M. to 4:00 P.M. when daylight saving time is in effect). Outside of these hours, emergency calls are those which affect your ability to control or view a process will be received by an answering service, and returned within one hour. The Solution Support Center maintains its own TDC 3000^X system, and frequently can duplicate problems on this equipment.

Time saving tip

It is a good idea to make specific notes about the problem before making the call. This will help to reduce delays and expedite answers.

5. Coprocessor Configuration Tasks

5.1 Configuration Overview

LCN requirements

The Universal Station^X requires software Release 430 or later. The Universal Station^X does not place any new requirements on LCN software or hardware. Approach the Universal Station^X planning and implementation tasks as you would for a standard Universal Station. All the LCN conventions for Universal Station implementation apply to the Universal Station^X. These include: node addressing, HM volume allotments, and NCF changes.

The Universal Station^X uses all LCN schematics/overlays without any data conversion. The Universal Station and Universal Station^X are equivalents as TDC 3000^X Process Control Computers. The Universal Station^X can load the standard personalities (UNP, ENG, and OPR) from the History Module or Removable Media, and also can load the UNPX personality from the coprocessor hard drive.

When loading the UNPX from the coprocessor hard drive, the button overlays, abstracts, schematics, and NCF are all read from the HM. The coprocessor hard drive stores only the QLTs and UNPX files for the K2LCN processor. The additional files cannot be stored on the coprocessor hard drive and must be loaded from the HM or Removable Media.

Currently, the UNPX personality is the functional equivalent of the UNP stored on the History Module; however, pending UNPX enhancements, it may provide functionality to the Universal Station^X that would not be available at the US.

NCF changes

Several fields were added to the NCF (Network Configuration File) to support Release 430. The NCF must be modified to reflect the Universal Station^X installation. This is done by following the standard NCF modification instructions available in the R430 Release Guide. The NCF must reflect that a Universal Station^X is present at the LCN node address pinned on your Universal Station^X. LCN node configuration, and console assignment must be completed prior to applying power to the station.

The NCF has the following changes: new personality type—Universal Station^X, new board type—TPDG, new board type—WSI2.

The SMCC (System Maintenance Control Center) has the following change: new board type—TPDG.

Coprocessor configuration

The major difference when compared to a Universal Station is the Universal Station^X's ability to display non-TDC 3000^X applications, generated by LAN (Local Area Network - Ethernet) computing resources. The TPDG board works in tandem with the coprocessor, providing the TDC display as an X client, which then is displayed through the X window/Motif environment. There are several attributes new to the TDC display, which are necessitated by viewing in the X window environment.

Installation complete?

This procedure assumes the Universal Station^X equipment to be properly installed and ready to apply power.

The following items must be completed, prior to the initial power on/login:

- Uncrating/positioning of the Universal Station^X equipment.
- Ground connections (power, cabinet, and Master Reference Ground)
- LCN node address pinning
- LCN cable connections
- LAN interface cable and transceiver attached to the Ethernet network
- Universal Station^X keyboard connection
- Universal Station^X mouse/trackball connection
- LCN printer connection (optional equipment)
- Laser printer connection (optional equipment)
- Visual inspection of all boards/cables to ensure proper seating
- Power Connection

Required information

Make a copy of the following lists for reference when performing the configuration procedures.

The information required to complete this process is listed below.

Desired hostname for the Universal Station^X _____
 Desired IP address _____
 Time/Date/Timezone _____
 Multiple network routers (Yes/No) _____
 (Single router only) IP address of router _____

Remote computer system information can be input in one of two ways:

1. Hostname and IP address for each when entered manually.

<i>hostname</i>	<i>###.###.###.###</i>
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____

Coprocessor Configuration Tasks – Configuration Overview

2. Source of existing `/etc/hosts` file when using ftp.
- IP address of source X _____
 - Login account at source _____
 - Password for account _____
 - Multiple network routers (Yes/No) _____
 - (Single router only) IP address of router _____

Coprocessor laser printer? **(Local/Remote/Unused)**

If Remote - Network information needed

- Hostname of print server _____
- IP address of print server _____
- Printer name _____

Coprocessor user information

Username/Access Group johndoe/Engineer

Shutdown WSI access

username

Native mode access

username

X hosts to be accessed

hostnames

Desired coprocessor password

⇒ **Note:** This password is extremely important. It is the key to your coprocessor security. Secure this document and control its access after completing the initial configuration.

Configuration checklist

Use this checklist to plan and record the various configuration tasks.

Step	Initials	Step performed	Subsection	Approx. time
1		Ensure installation complete? _____ Uncrating/positioning of the Universal Station ^X equipment. _____ Ground connections (power, cabinet, and Master Reference Ground) _____ LCN node address pinning _____ LCN cable connections _____ LAN interface cable and transceiver attached to the Ethernet network _____ Universal Station ^X keyboard connection _____ Universal Station ^X mouse/trackball connection _____ LCN printer connection (optional equipment) _____ Laser printer connection (optional equipment) _____ Visual inspection of all boards/cables to ensure proper seating _____ Power connection _____ Information prepared	N/A	1 hour
2		Initial power on completed?	5.2	3-5 minutes
3		Initial login completed?	5.3	1 minute
4		Coprocessor password configured?	5.4	5 minutes
5		Time/date/timezone entered?	5.5	2 minutes
6		Station coprocessor hostname and Internet Protocol address entered?	5.6	5 minutes
7		Remote hostnames and addresses entered?	5.7	4-10 minutes
8		Network routing configured?	5.8	4-10 minutes
9		Coprocessor printers configured?	5.10	15-20 minutes
10		Shutdown WSI access configured?	5.11	2-30 minutes
11		Native mode access configured?	5.11	2-30 minutes

5.2 Initial Power-On

Apply power

Apply power using the following steps:

Step	Action
1	Open front access door to gain access to the circuit breaker.
2	Place the circuit breaker in the ON position.
3	Open the rear access door to gain access to the 5-slot module.
4	Apply power to the station using the power switch on the 5-slot module faceplate.
5	Observe the K2LCN board LEDs during the power on self-test. If no errors are detected, the appropriate LCN node address is displayed.
6	<u>If the appropriate LCN node address is displayed</u> Proceed with Step 7. <u>If another number is displayed</u> Record the error number and contact maintenance personnel to correct the fault. Proceed with Step 9.
7	Wait 5 minutes for the Login banner to appear.
8	<u>If the Login banner does appear</u> Proceed with subsection 5.3, <i>Initial Login</i>. <u>If the Login banner has not appeared</u> Proceed with Step 9.
9	Using the power switch located on the 5-slot module faceplate, remove power from the 5-slot module.
10	Set the station circuit breaker to the OFF position. Contact maintenance to correct the fault.

5.3 Initial Login

Preconfigured users

The Universal Station^X coprocessor environment is preconfigured with one user in each of the four groups. The initial login passwords of the preconfigured users should be changed to ensure security.

The steps for changing the preconfigured user passwords are presented following the initial configuration procedures covered in this section.

Coprocessor (root) password

The coprocessor itself must be password protected; however, to ease installation it is shipped without a password. During initial installation procedures, the coprocessor has no password. When asked to type the password, press <RETURN>.

Only users with Engineer Access rights have configuration access to the coprocessor environment. Prior to obtaining access, however, the user must provide the coprocessor (root) password.

The steps for changing the coprocessor (root) password is presented following the initial configuration procedures.

Initial login passwords

The preconfigured user names and passwords are as shown:

User Name	Password	Coprocessor Access
engineer	engineer	Engineer Access
operator	operator	Operator Access
supervisor	supervisor	Supervisor Access
viewOnly	viewOnly	View-Only Access

Login

Login as the user “engineer,” using the steps below:

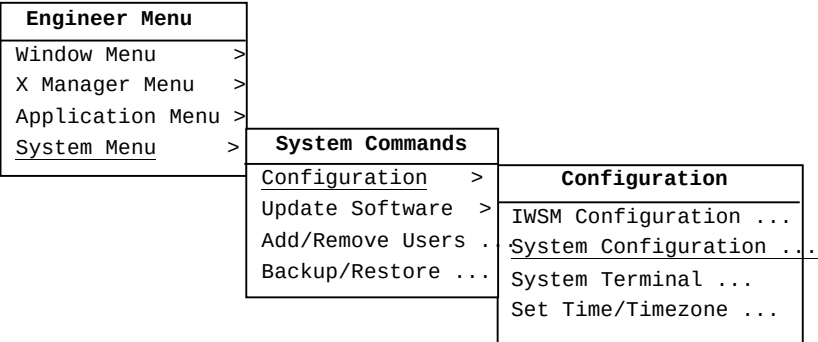
Step	Action
1	Position the cursor in the Login box and type engineer <RETURN> Note: The password box will be selected. Do not move the mouse/trackball. If the cursor is moved, it will be necessary to reposition the cursor over the password box.
2	Type the password engineer <RETURN>.

5.4 Change Coprocessor (root) Password

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is required.

Menu appearance



51143

Change the coprocessor (root) password

Changing the coprocessor (root) password to ensure security.

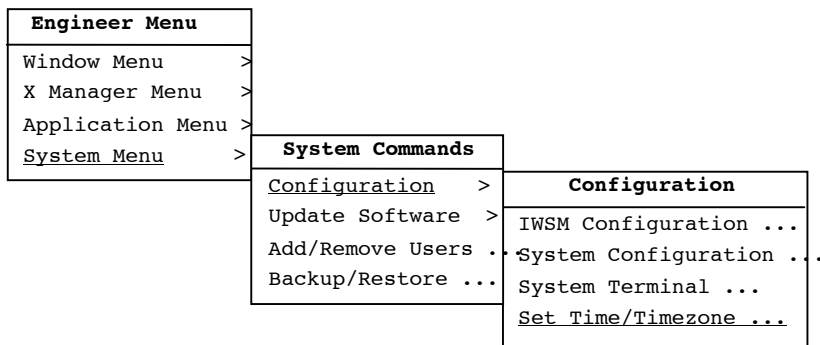
Step	Action
1	Select System Menu > Configuration > System Configuration from the workspace menu .
2	Type the coprocessor (root) password and <RETURN> at the prompt. Note: Only <RETURN> is necessary on a new station. The System Administration Manager (SAM) window appears.
3	Click on Accounts for User & Groups and select Actions > Open Item.
4	Click on Users and select Actions > Run.
5	Select the line of the display containing the login name of "root".
6	Select Actions > Modify.
7	Select Yes if you are prompted to confirm this action.
8	Select the Change Password target located in the lower left corner of the window.
9	Enter the new password<RETURN> .
10	Re-enter the new password<RETURN> .
11	Select ok .
12	Select ok .
13	Select File > Exit from the top of the Users and Groups window.

5.5 Configure Coprocessor Time and Timezone

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is required.

Menu appearance



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Set timezone

Set the timezone using the following procedure.

Step	Action
1	Select System Menu > Configuration > Set Time/Timezone from the workspace menu.
2	Type the coprocessor (root) password and <RETURN> at the prompt. The Timezone window appears.
3	Make the appropriate responses within a series of drop down menus and then click OK when finished.
4	Click Yes when the Confirm Time Zone prompt appears.
5	A Reboot System menu appears with the following message: "You must reboot the system to make your changes take effect. Would you like to reboot the system now?" Select one of the following: • Yes, Reboot • No, Do not Reboot

Set date and time

Set the date and time using the following procedure.

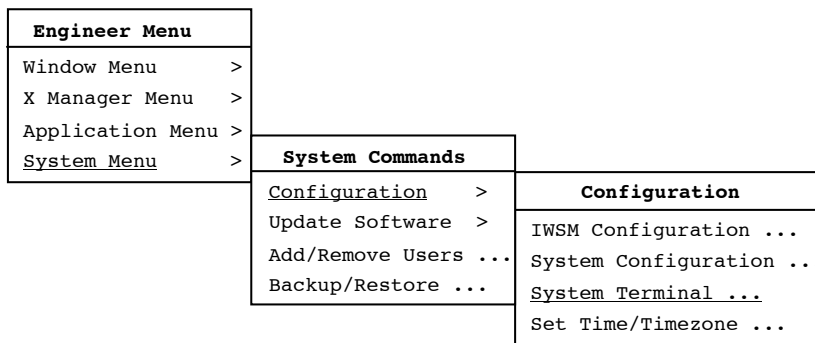
Step	Action
1	Select System Menu > Configuration > System Configuration from the workspace menu.
2	Type the coprocessor (root) password and <RETURN> at the prompt. The System Administration Manager (SAM) window appears.
3	Click the Time icon and select Actions > Open Item .
4	Click System Clock and select Actions > Run . A Set Time window appears.
5	Enter the correct time and date and click OK.
6	Select File > Exit SAM .
7	It is a good idea to reboot the system after entering date and time. This will ensure that the system uses the correct time and date for time dependent functions such as cron, and file time stamps. Note: If you are setting the time as part of an initial coprocessor setup, you can defer the reboot process until after the coprocessor node name and IP address procedure.

5.6 Configure Coprocessor Node Name and IP Address

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is required.

Menu appearance



51140

Procedure overview

A special utility called “set_parms” is used on a U^XS station to set up the coprocessor host name and the internet protocol (IP) address. Be prepared to supply the host name and IP address when prompted. The coprocessor must be rebooted to enable the change.

Set hostname and IP address

Step	Action
1	Select System Menu > Configuration > System Terminal from the Workspace Menu.
2	Enter the coprocessor (root) password <RETURN> . Note: Only <RETURN> is necessary on a new station.
3	At the prompt enter <code>/etc/set_parms hostname<RETURN></code> . Note: Enter the characters <code>h o s t n a m e</code> . Do not enter the hostname of your station here. The <code>set_parms</code> program will prompt you to enter the hostname of your station.
4	Respond to the prompts using appropriate keyboard entries for your configuration. Use <RETURN> to enter values. Note: When asked if you want to reboot now, enter <code>n</code> and press <RETURN> . You will reboot the station later in this procedure.
5	After returning the system terminal window prompt, you must now configure the station coprocessor hostname for DAT tape backups. Enter <code>cd /usr/lib/X11<RETURN></code> .

Coprocessor Configuration Tasks – Configure Coprocessor Node Name and IP Address

Step	Action
6	Edit the <code>system.Xdefaults</code> file. Enter <code>vi system.Xdefaults<RETURN>.</code>
7	Search the file for the first instance of <code><host></code> . Enter <code>/<host <RETURN>.</code>
8	Change <code><host name></code> to the actual hostname of your station. Enter <code>:s/<host name>/{coprocessor name}<RETURN>.</code> Note: Substitute <code>{coprocessor name}</code> with the actual name of your station as entered earlier in this procedure. Do not include the curly brackets in your command. Example: <code>:s/<host name>/trn2 <RETURN>.</code>
9	Un-comment this line. Position the cursor on the <code>!</code> at the beginning of the line. Type <code>x</code> to remove it.
10	Save the file and quit the editor. Enter <code>:wq!<RETURN>.</code>
11	Enter <code>reboot -q<RETURN></code> Note: This will cause the coprocessor to reboot . This enables the new hostname and IP address. Do not touch the red target that appears during reboot.

5.7 Configure Coprocessor LAN Information

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is required.

Menu appearance

Engineer Menu	
Window Menu	>
X Manager Menu	>
Application Menu	>
<u>S</u> ystem Menu	>

System Commands	
<u>C</u> onfiguration	>
Update Software	>
Add/Remove Users	..
Backup/Restore	...

Configuration	
IWSM Configuration	...
<u>S</u> ystem Configuration	...
System Terminal	...
Set Time/Timezone	...

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Set remote hostnames and TCP/IP addresses

Setup remote hostnames, and their TCP/IP addresses using the following procedure.

Step	Action
1	Select System Menu > Configuration > System Configuration from the workspace menu.
2	Type the coprocessor (root) password and <RETURN> at the prompt. The System Administration Manager (SAM) window appears.
3	Click Networking and Communications and select Actions > Open Item .
4	Click Internet Addresses and select Actions > Run . SAM displays the remote system names and Internet addresses that are already configured.
5	Select Actions > Add . The Add Internet Address window appears.
6	Enter the IP address of the remote system. Upon exiting the Internet Address field, SAM checks to make sure you have entered a valid IP/Internet address. SAM also determines if a gateway is required for the connection.
7	Enter the remote system name . Note: Hostnames can consist of a maximum of 64 alphanumeric, underscore, and period characters. Upon exiting the Remote System Name field, SAM checks to make sure that connectivity has not already been configured for this system. If it has, SAM displays an error message.
8	Select OK.
9	Repeat steps 5 – 8 as necessary.
10	Select File > Exit .
11	Select Exit SAM .

5.8 Configure LAN Routing Information

Routing choices

The Universal Station^X coprocessor may be configured for either static or dynamic routing of LAN information. The Universal Station^X is configured by default for static routing. Typically the routing configuration will be supplied by a network support individual responsible for the Management of the LAN.

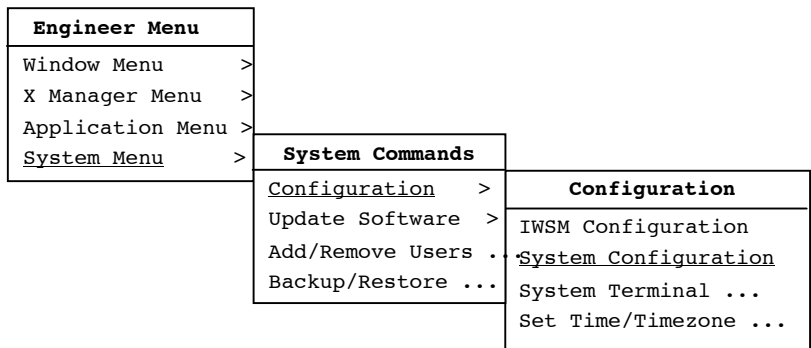
If the LAN has no routers or only one router, the gated daemon should not be used and the hostname and TCP/IP address for the default gateway should be configured.

If the LAN uses multiple routers and hosts which are configured with subnet masks, it will be necessary to configure and use the gated daemon. The coprocessor will interrogate the routers and use the routers dynamic routing table information. Configuring the gated daemon process is covered in the Hewlett Packard document “*Installing and Administering ARPA Services.*”

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is required.

Menu appearance



51143

Setup static LAN routing

⇒ **Note:** You must complete the “Configure Coprocessor Node Name and IP Address” procedure before you perform this procedure:

The default gateway can be set using the following procedure:

Step	Action
1	Select System Menu > Configuration > System Configuration from the workspace menu.
2	Type the coprocessor (root) password and <RETURN> at the prompt. The System Administration Manager (SAM) window appears.
3	Click Networking and Communications and select Actions > Open Item .
4	Click Internet Addresses and select Actions > Run . SAM displays the remote system names and Internet addresses that are already configured.
5	Select Actions > Add . The Add Internet Address window appears.
6	Enter the IP address of the default gateway node.
7	Enter the hostname of the default gateway node and select OK.
8	Select Actions > Specify Default Gateway .
9	Enter the default gateway internet address and select OK.
10	Select File > Exit .
11	Select Exit SAM .

5.9 Configure Security for Remote Users

Overview

The default configuration of your Release 300 system does not allow any remote users to connect to the station. The file `inetd.sec`, which is located in the `/usr/adm` directory, is used for this purpose. Refer the Hewlett-Packard manual titled “*Using ARPA Services*” for a detailed discussion of this file.

WARNING

The `inetd.sec` file may be overwritten during a coprocessor software update to the U^XS system. The system administrator should verify the accuracy of this file as part of the upgrade process.

Configuration

The default state of the `inetd.sec` file is set to “deny” for all entries. The file should be modified using the HP-UX SAM utility to allow remote access for various system services.

Step	Action
1	Select System Menu > Configuration > System Configuration from the workspace menu.
2	Type the coprocessor (root) password and <RETURN> at the prompt. The System Administration Manager (SAM) window appears
3	Click on Networking and Communications and select Actions > Open Item .
4	Click on System Access and select Actions > Open Item .
5	Click on Internet Services and select Actions > Run .
6	Select Actions > Modify All Services . WARNING This enables ALL services for ALL devices anywhere on the network. This can be a security risk. Instead, you may want to consider the following alternative: • Select the service you wish to modify (enable). • Select Actions > Modify. • Set the System Permission as appropriate. Click OK
7	Select Exit SAM .

5.10 Configure Coprocessor Printer (Network)

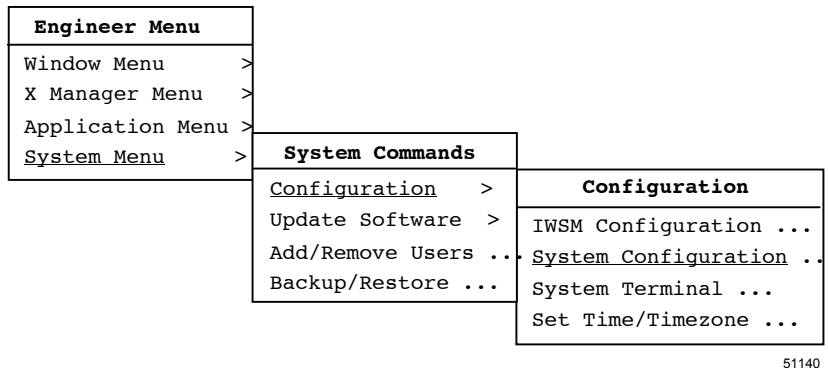
Local printer install

This procedure will configure a coprocessor network printer. The printer will be available to be configured on other Universal Station^X stations.

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is required.

Menu appearance



Configure network printer

Configure a coprocessor network printer using the following procedure:

Step	Action
1	Select System Menu > Configuration > System Configuration from the workspace menu
2	Type the coprocessor (root) password and click OK. The System Administration Manager (SAM) window appears
3	Click on Printers and Plotters and select Actions > Open Item .
4	Click on LP Spooler and select Actions > Open Item .
5	Click on Printers and Plotters and select Actions > Run .
6	Select Actions > Add Remote Printer/Plotter
7	- Click on each of the following buttons within the Add Remote Printer/Plotter window and enter the appropriate information for each option: - Printer Name: - Remote System Name: - Remote Printer Name: - Click OK when finished.
8	Select Actions > Set as System Default Destination

Step	Action
9	Select File > Exit .
10	Select File > Exit SAM .

JetAdmin Utility

HP JetAdmin is a software utility that allows system administrators to configure and manage print servers and network printers and to share printers in LAN environments. JetAdmin has been updated for the HP-UX 10.20 operating system. The software files, now in the *swinstall* file format, are available at the HP Support website.

Updating JetAdmin for HP-UX 10.20

If you wish to update the JetAdmin utility for the 10.20 operating system, you can download the necessary files. The following table contains a list of JetAdmin software and documentation files.

JetAdmin file locations and information

File Name	Description	Available At...	URL path
HP10d615.SD	Swinstall format file of hpnp JetAdmin Software version D.06.15 for the HP-UX 10.0, 10.01, 10.20 operating system on an HP9000 Series	http://www.hp.com/ , Services and Support, Software Support, Print Drivers, Network Printing, HP JetAdmin Software for HP-UX 10.x, Software and Drivers	http://www.hp.com/cposupport/networking/software/ja240en.sd.html
INSTALL.UNX	Readme file for installing HP JetAdmin on HP-UX. This file contains complete installation instructions of the JetAdmin utility using swinstall. This file also tells how to add jetprint with Jetadmin on HP-UX 10.x.	http://www.hp.com/ , Services and Support, Software Support, Print Drivers, Network Printing, HP JetAdmin Software for HP-UX 10.x, HP Customer Care Search, Type INSTALL.UNX and click GO, Select HP JetAdmin for HP-UX 10.x on HP9000 series 700 or 800 , Go to the Download Section	ftp://ftp.hp.com/pub/networking/software/INSTALL.UNX
README.UNX	Technical support information	http://www.hp.com/ , Services and Support, Software Support, Print Drivers, Network Printing, HP JetAdmin Software for HP-UX 10.x, HP Customer Care Search, Type INSTALL.UNX and click GO, Select HP JetAdmin for HP-UX 10.x on HP9000 series 700 or 800 , Go to the Download Section	ftp://ftp.hp.com/pub/networking/software/README.UNX
README0.UNX	Index/description of HP JetAdmin for UNIX files.	http://www.hp.com/ , Services and Support, Software Support, Print Drivers, Network Printing, HP JetAdmin Software for HP-UX 10.x, HP Customer Care Search, Type README0.UNX and click GO, Select HP JetAdmin for HP-UX 10.x on HP9000 series 700 or 800 , Go to the Download Section	ftp://ftp.hp.com/pub/networking/software/README0.UNX
bpj05461.pdf	HP JetDirect Print Server Software Installation Guide: Chapter 1, "Introducing the HP JetDirect Print Server Software" Chapter 10, "UNIX (HP-UX and Solaris) Networks"	http://www.hp.com/ , Services and Support, Software Support, Print Drivers, Network Printing, HP JetAdmin Software for HP-UX 10.x, Product Manuals	http://www.hp.com/cposupport/manindex/hpjetadmin1873_eng_man.html
Assorted Support Documents	Support documents for troubleshooting purposes and FAQ's.	http://www.hp.com/ , Services and Support, Software Support, Print Drivers, Network Printing, HP JetAdmin Software for HP-UX 10.x, Support Documents for Installing, Configuring and Using	http://www.hp.com/cposupport/docindex/hpjetadmin1873_installingconfigu.html

5.11 Industrial Workspace Manager (IWSM) Configuration

Required access level

The user must have engineer group access and perform these procedures while logged in as “engineer.” The coprocessor (root) password is required.

Menu appearance

Engineer Menu	
Window Menu	>
X Manager Menu	>
Application Menu	>
System Menu	>

System Commands	
Configuration	>
Update Software	>
Add/Remove Users...	
Backup/Restore...	

Configuration	
IWSM Configuration ...	
System Configuration ...	
Set Time/Timezone ...	

51151

IWSM configuration tool

◇ Enter Native Mode

◆ Shutdown the WSI

ALL	root
ALL	operator
ALL	engineer
ALL	username

Where *username* is replaced with *user_login_name*

Add Delete Close

Display Area Lists
Co-processor Users
Allowed to Perform the
Action Selected Above

Text Entry Area

51152

Shutdown WSI authorization

Use the following procedure to authorize specific users to shutdown the coprocessor:

Step	Action
1	Select System Menu > Configuration > IWSM Configuration from the workspace menu.
2	Type the coprocessor (root) password and <RETURN> at the prompt.
3	Select the Shutdown the WSI target.
4	Select the text entry area of the IWSM Configuration Tool. (Just above the Add, Delete, and Close targets).
5	Type information using this syntax: <code>*ALL* username</code> where <i>username</i> = user's login name Important: Use spaces to separate <code>*ALL*</code> from the user name.
6	Click on the Add target to authorize the user to shut down the WSI.
7	Repeat Steps 4 through 6 for each user requiring shutdown privileges.
8	Select the Close target.

Remove Shutdown WSI authorization

Use the following procedure to remove authorization to shut down the coprocessor from specific users:

Step	Action
1	Select IWSM Configuration from the workspace menu.
2	Type the coprocessor (root) password and <RETURN> at the prompt.
3	Select the Shutdown the WSI target.
4	Select the user name to be deleted.
5	Select the Delete target.
6	Repeat Steps 4 and 5 for all users to be deleted.
7	Select the Close target.

6. Coprocessor User Environment Configuration

6.1 User Configuration Overview

Coprocessor configuration

The major difference, when compared to a Universal Station, is the Universal Station^X's ability to display non-TDC 3000^X applications, generated by LAN (Local Area Network - Ethernet) computing resources.

There are several attributes new to the Universal Station^X display, which are necessitated by viewing in the X Windows environment:

- X Host Authorization—Controls remote access to the Universal Station^X display
- Native Window Behavior—Controls dominance of the LCN display
- Keyboard Focus Policy—Controls keyboard/mouse pointer interaction

Custom configuration

The Universal Station^X coprocessor configuration can be customized to alter the standard user environment. This could include:

- Providing custom workspace menu choices
- Altering shell variables (path, terminal, prompt, etc.)
- Automatically mounting remote filesystems (HP LaserRom)
- Automatically running applications at login

Customizing the files which control the coprocessor environment should be done prior to registering users. When the customization is completed, the users can be registered and additional systems loaded with the custom configuration.

Required information

The following information is required for configuration:

- The hostnames of any remote computing resources which will open an X display on this station. This access is granted on a user-by-user basis using subsection 6.2, "X Host Authorization."

6.2 X Host Authorization

What is it?

The X host authorization application grants X display permission to remote systems on an individual-user basis. This allows other computer systems to open a window on the user's Universal Station^X display.

Required access level

The procedures can be performed by any authorized user. This authorization is granted/recorded on an individual-user basis.

Menu appearance

Engineer Menu	
Window Menu	>
X Manager Menu	>
Application Menu	>
System Menu	>

X Manager Menu
<u>X Host Authorization</u>
Change User Password
Refresh
Restart ...
Logout ...

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X host addresses

To authorize remote systems to display X Window applications on the Universal Station^X, the hostname and TCP/IP address must be entered prior to this step, using the LAN information entry procedure.

X Host tool

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X host authorization scope

- The user can choose from two methods of granting access permission:
- Explicit host authorization—user must enter the hostnames of all systems requiring access
- Grant access to all hosts—user selects Any host is allowed access

Explicit host authorization

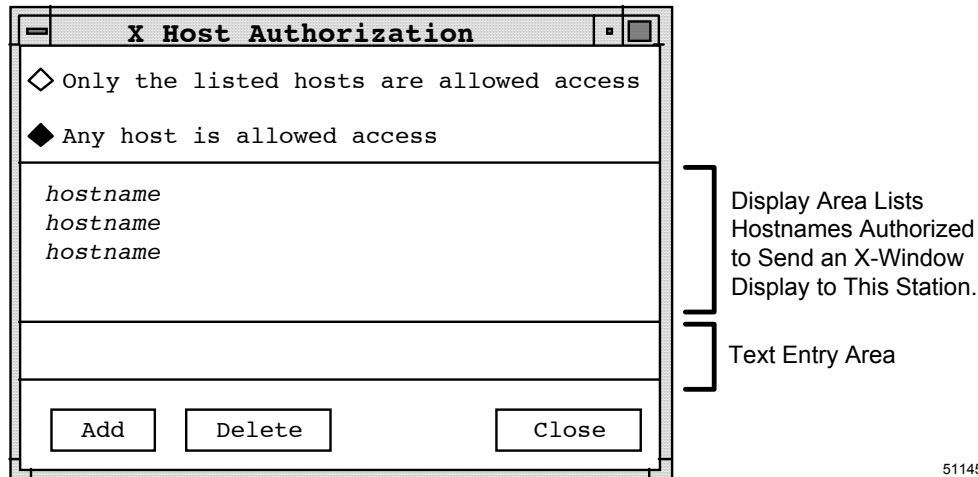
Use the following to authorize specific X hosts access:

Step	Action
1	Select X Host Authorization from the workspace menu.
2	Type the coprocessor (root) password and <RETURN> at the prompt.
3	Select Only the listed hosts are allowed access target.
4	Click on the text entry area of the IWSM Configuration Tool.
5	Type information using this syntax: hostname hostname hostname hostname hostname = replace with a valid hostname Use of spaces is not allowed. The tool does nothing if there are invalid characters in the text entry field.
6	Click on the Add target to authorize the remote host to display locally.
7	Repeat Steps 4 through 6 for each host requiring X host access.
8	To delete a host's X display authorization privileges, click on the hostname to be deleted.
9	Click on the Delete target.
10	Repeat Steps 8 and 9 for each host to be deleted.
11	Select the Close target.

Allowing all X-hosts access

All LAN computing resources listed in the hosts file can be authorized to display X-window applications on this station using the following procedure.

Step	Action
1	Select X Host Authorization from the workspace menu.
2	Type the coprocessor (root) password and <RETURN> at the prompt.
3	Select the Any host is allowed access target.
4	Select the Close target.

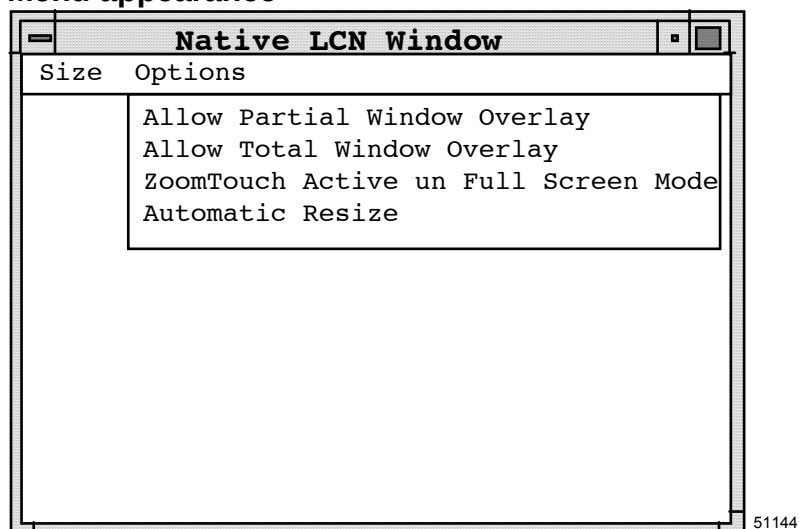


6.3 Configure Native Window Behavior

Window dominance

The dominance of the Native LCN Window is configurable on a user-by-user basis. The attribute allows the Native LCN Window to be partially obscured by another window. The Native LCN Window behavior is initially configured to allow only engineering group members to obscure the LCN display. However, the individual users (all access levels) can set this attribute at their discretion.

Menu appearance



Allow/Disallow partial window overlay

The selection of this option toggles on/off. Therefore, perform the same steps to allow partial window overlay and to disallow partial window overlay.

Step	Action
1	Select Options from the Native LCN Window menu bar. Note: If a square appears to the left of the Allow Partial Window Overlay option, it indicates that the option is currently allowed. This will allow the Native LCN Window to be partially obscured by another window. If no square appears to the left of the Allow Partial Window Overlay option, it indicates that the option is currently disallowed. This will not allow the Native LCN Window to be partially obscured by another window.
2	Select Allow Partial Window Overlay to change the option. Note: The change will take effect immediately.

Allow/Disallow total window overlay

The selection of this option toggles on/off. Therefore, perform the same steps to allow total window overlay and to disallow total window overlay.

Step	Action
1	Select Options from the Native LCN Window menu bar. Note: If a square appears to the left of the Allow Total Window Overlay option, it indicates that the option is currently allowed. This will allow the Native LCN Window to be totally obscured by another window. If no square appears to the left of the Allow Total Window Overlay option, it indicates that the option is currently disallowed. This will not allow the Native LCN Window to be totally obscured by another window.
2	Select Allow Total Window Overlay to change the option. Note: The change will take effect immediately.

Setting default native window behavior

Default behavior for the Native LCN Window during future login sessions can be configured for each user. This is accomplished by modifying the `.xdefaults` file in the individual user's directory.

To set up the Native LCN window in the `operator1` account to allow other windows to obscure it (partially or totally), change (or add) the following lines in the `.xdefaults` file in `operator1`'s directory (`/users/XUS/operator/operator1`).

```
NativeLCNWindow*allowTotalObscurity:      False
NativeLCNWindow*allowPartialObscurity:    True
```

To set up the Native LCN window in the `operator1` account to automatically resize the LCN display to the current window size, change (or add) the following line in the `.xdefaults` file in `operator1`'s directory (`/users/XUS/operator/operator1`).

```
NativeLCNWindow*autoZoom:                  True
```

Global Native LCN window behavior for all users can be achieved by changing the lines above in the `NativeLCNWindow` file in the `/usr/lib/x11/app-defaults` directory.

6.4 Configure Keyboard Focus Policy

Required access level

The procedure can be performed while logged in as any user.

Menu appearance

Operator Menu		
<u>Window Menu</u> >	Window Menu	
X Manager Menu >	Shuffle >	
Application Menu >	Other Screen	
	<u>Keyboard Focus Policy</u> >	Explicit
	Window Print	Pointer

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Explicit vs. Pointer

The keyboard focus policy allows two choices:

- Explicit focus—user must click on a window to make it active
 - Pointer focus—user must point at the window to make it active
- ⇒ **Note:** Even small mouse movements may inactivate the window when using pointer focus. This may cause keyboard entries to be lost or directed to the wrong window.

Explicit focus

Use the following to set the focus policy to explicit:

Step	Action
1	Select Keyboard Focus Policy – Explicit from the workspace menu. Note: The change in focus policy will not take effect until the next login.

Pointer focus

Use the following to set the focus policy to pointer:

Step	Action
1	Select Keyboard Focus Policy – Pointer from the workspace menu. Note: The change in focus policy will not take effect until the next login.

6.5 Change User Password

Required access level

The procedure can be performed while logged in as any user. The current password is required.

Menu appearance

Operator Menu			
Window Menu	>		
<u>X Manager Menu</u>	>		
Application Menu	>		
		X Manager Menu	
		X Host Authorization	
		<u>Change User Password</u>	
		Refresh	
		Restart ...	
		Logout ...	

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Change the user password

Change the user password frequently to ensure security.

Step	Action
1	Select Change User Password from the workspace menu.
2	Type the old password and <RETURN> at the prompt. Note: The new password must be at least 6 characters, containing at least 2 alpha characters and 1 numeric character.
3	Type a new password and <RETURN>.
4	Retype the new password and <RETURN> to confirm the entry.

6.6 Customizing the Workspace Menu

Why customize?

The workspace menu can contain items which connect to and run application software on remote systems and cause them to display locally. This reduces the steps and knowledge required to perform an operational task.

Modify all users

By modifying the file in the appropriate `.baseUser` directory, the customization can be for all subsequently registered users of a specific group. Remember to transfer this file to the `.baseUser` directory on all machines.

To use the new custom menu, prior users must have the new file transferred to their home directory.

Modify a single user

By modifying the `.mwmrc` file in the user's home directory, only that user will have the altered workspace menu. Remember to transfer this file to the user's home directory on all machines where he is registered.

Appropriate files

A customized command string can be appended to the workspace menu by placing the command string into the appropriate `.mwmrc` file.

- `/users/XUS/operator/.baseUser/.mwmrc`
- `/users/XUS/operator/username/.mwmrc`
- `/users/XUS/supervisor/.baseUser/.mwmrc`
- `/users/XUS/supervisor/username/.mwmrc`
- `/users/XUS/engineer/.baseUser/.mwmrc`
- `/users/XUS/engineer/username/.mwmrc`
- `/users/XUS/viewOnly/.baseUser/.mwmrc`
- `/users/XUS/viewOnly/username/.mwmrc`

Parts of a command string

There are several components of a workspace menu command string to be placed into the `.mwmrc` file. The elements of the string are the same for all remote computing systems, including those with dissimilar operating systems). However, the commands and syntax of the remote system are used in one element of the command string. This requires knowledge of the remote operating systems commands and operation. This knowledge will not be covered in this manual and can be located in the documentation of the appropriate system.

The elements of the workspace menu command strings:

MENU ITEM NAMES		WINDOW MANAGER COMMANDS
Menu RootMenu		
{		
"Engineer Menu"	f.title	
"X Menu"	f.menu	XMenu
"Application Menu"	f.menu	ApplicationMenu
"System Menu"	f.menu	SystemMenu
}		
Menu XMenu		
{		
"X Menu"	f.title	
"Shuffle"	f.menu	ShuffleMenu
"Logout"	f.quit_mwm	
}		
Menu ApplicationMenu		
{		
"Applications"	f.title	
"Unix Load"	f.exec	"xload-geom -4+4 &"
"Calculator"	f.exec	"xcalc -geom -160+4 &"
"Datebook"	f.exec	"datebook -geom +724+4 &"
"Telnet"	f.exec	"xterm -geom +291+400 -e telnet &"
"DecNet"	f.exec	"xteemx340 -geom +291+400 &"
}		

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Creating a command

To create the command line, use the vi text editor to modify the appropriate `.mwmrc` file. Here are the steps to creating a command string to be included in the workspace menu.

Step	Action
1	Choose a menu label.
2	Choose the appropriate window manager command.
3	Determine if a remote shell is necessary. It is required to execute applications on remote computers.
4	Choose the system command including any additional parameters as appropriate to the software application.

Choosing a label

Choose a label name that represents the action as close as possible. Time spent planning will provide a menu system that is intuitive and requires little (if any) additional documentation.

Choose a window manager command

Several window manager commands are used to customize the workspace menu and are listed here for your convenience. For additional information, see Hewlett Packard publication, *Using the X Window System*.

Category	Command	Description
Program Execution	<code>f.exec "cmd &"</code>	Executes <i>cmd</i> using <code>/bin/sh</code>
	<code>f.restart</code>	Restarts the Motif Window Manager
	<code>f.kill</code>	Kills the client that started the window
Menu Operations	<code>f.menu MenuName</code>	Associates a menu with a selection
	<code>f.post_wmenu</code>	Posts a window menu
	<code>f.separator</code>	Draws a line between selections
	<code>f.title</code>	Inserts a title into menu at specified position
	<code>f.nop</code>	Disable a menu entry with no operation

Local commands

To execute commands on the local system, only the window manager command and system command are necessary.

Required syntax for executing local applications

```
"MenuItemName"      f.exec "ApplicationName &"
```

```
"MenuItemName"      f.exec "ScriptFileName &"
```

An example of the command string to execute a local application.

```
"Datebook"          f.exec "datebook &"
```

Remote commands

To execute commands in a remote computer system, a remote shell session is needed. Add the following to the window manager command to build the command string.

Required syntax for executing remote applications

```
"MenuItemName"      f.exec "remsh hostname ApplicationName &"
```

```
"MenuItemName"      f.exec "remsh hostname ScriptFileName &"
```

An example of the command string to execute the application Framemaker on the host system named hp700_01.

```
"Word Processor"    f.exec "remsh hp700_01 maker &"
```

Sample customization

The following is an example of the existing Application Menu customized to provide an automatic connection to remote computers running various software. It is an excerpt of the standard `.mwmrc` file with additional lines (shown in larger print) to produce the workspace menu changes.

Menu ApplicationMenu

```
{
"Applications"      f.title
"Unix Load"        f.exec "xload -geom -4+4 &"
"Calculator"        f.exec "xcalc -geom -160+4 &"
"Datebook"         f.exec "datebook -geom +724+4 &"
"Telnet"           f.exec "xterm -geom +291+400 -e telnet &"
"DECnet"           f.exec "xteemx340 -geom +291+400 &"
"Lab Analysis"     f.menu LabMenu
}
```

Menu LabMenu

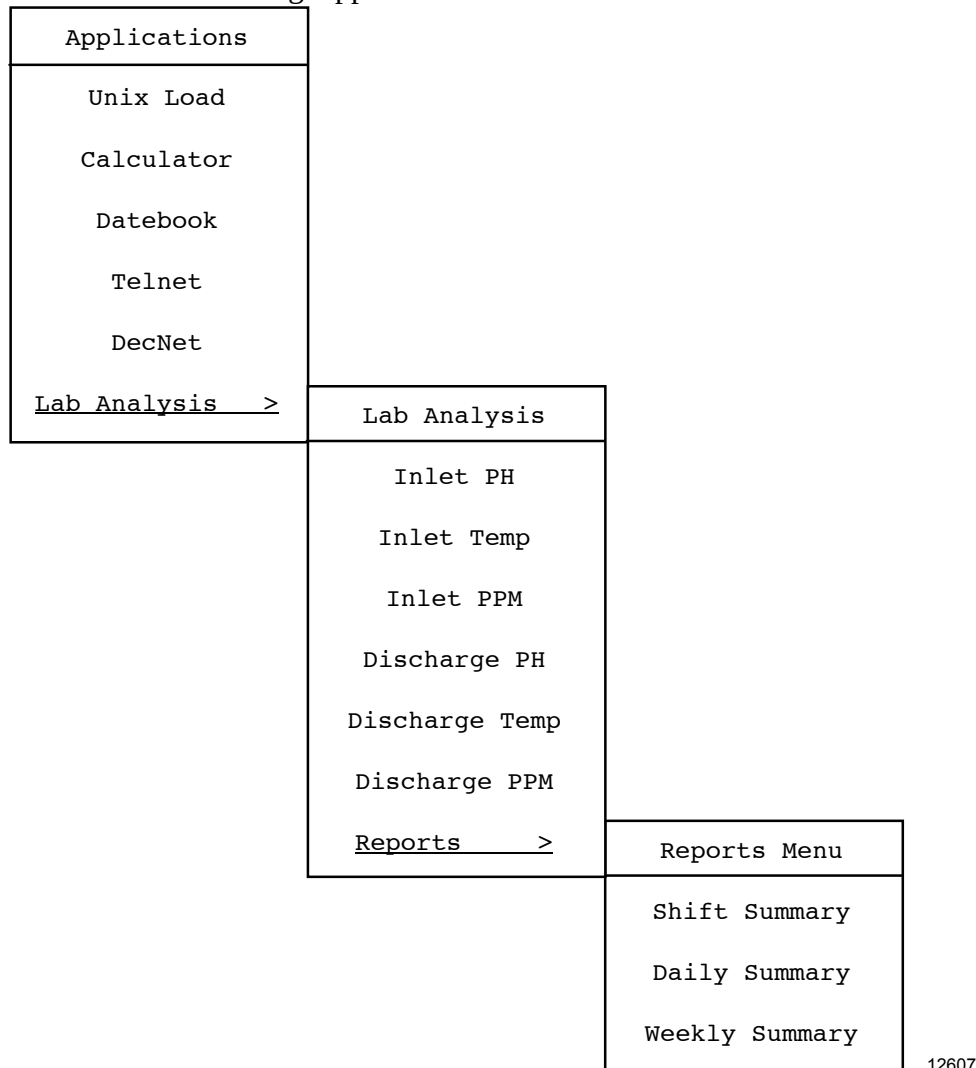
```
{
"Lab Analysis"      f.title
"Inlet PH"          f.exec "remsh hp700_01 -l Inlet_PH raw1 &"
"Inlet Temp"        f.exec "remsh hp700_02 -l Inlet_Temp raw2 &"
"Inlet PPM"         f.exec "remsh vax01 -l Inlet_Clarify raw3 &"
"Discharge PH"      f.exec "remsh hp700_2 -l Waste_PH out1 &"
"Discharge Temp"    f.exec "remsh hp700_2 -l Waste_Temp out2 &"
"Discharge PPM"     f.exec "remsh vax01 -l Waste_Clarify out3 &"
"Reports"           f.menu ReportMenu
}
```

Menu ReportMenu

```
"Reports"          f.title
"Shift Summary"     f.exec "RunShiftSum &"
"Daily Summary"     f.exec "RunDailySum &"
"Weekly Summary"    f.exec "RunWeekSum &"
}
```

Sample Customization, continued

Produces the following Applications Menu with two new sub menus:



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6.7 Customizing the User's Shell Variables

What is the shell?

The shell is the mechanism used to issue commands to the UNIX processor. This shell acts as a command interpreter and controls the UNIX system resources to carry out the requested task. The shell provides a powerful and flexible programming language to allow creation of script files containing many commands.

Available shells

There are several shells used

- `/bin/ksh` — Korn shell used for members of the group Engineer
- `/bin/rksh` — restricted Korn shell used for members of the groups: Operator, Supervisor, and View Only
- `/bin/csh` — C shell used for the root user

Shell variables

The shell tracks items called variables that it uses to control system resources.

- The most often used variables are
- `SHELL` - Pathname of your shell
- `TERM` - Defines your terminal type for screen oriented commands
- `PATH` - List of directories the shell searches for commands

Controlling files

The shell variables are set at login and each time a shell is created. The system executes various initialization files depending on the users shell type. These files are located in the users home directory.

Shell type	Controlling files
Korn shell — /bin/ksh	.profile — executed at login \$ENV — when a subshell is spawned Note: \$ENV is a variable that defines the name of the file. Commonly .kshrc
Restricted Korn shell — /bin/rksh	.profile — executed at login \$ENV — when a subshell is spawned
C shell — /bin/csh	.login — executed at login .cshrc —when a subshell is spawned

For more information

For more information, see the following Hewlett Packard publications:

- A Beginners Guide To Using Shells
- HP-UX Concepts and Tutorials: Shells and Miscellaneous Tools
- HP-UX System Security

7. Coprocessor Administration Tasks

7.1 Create New Coprocessor Users

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is not required.

Guidelines

Use the following criteria when adding a user to the coprocessor environment:

- Assign a separate login name for each user
 - Do not assign a login name beginning with a number
 - Determine the user’s group access level by providing the “least privileges” necessary to perform the required tasks
 - Do keep a record of all users assigned access, it should contain:
 - User name, contact information (phone number, address, etc.)
 - User login name
 - User access level
 - List all systems the user has accounts on
 - List all software where the user has accounts (E-mail, Telex, etc.)
- ⇒ **Note:** This listing should be maintained by the system administrator.

Trusted Systems

WARNING

Honeywell strongly recommends that you do not make your Universal Station^X s trusted systems

The coprocessor user environment is not intended to be used to configure a secure (trusted) system. If a trusted system configuration is required, use the HP-UX System Administration Manager (SAM) utility for this purpose. Refer to the manual “*HP-UX System Security*” for further information on setting up a trusted system.

.baseUser files

The .baseUser directory in each user category (engineer, operator, etc.) contains the default files that will be used to initially configure a new user’s own directory. Be aware that Honeywell may update these files during a software release upgrade procedure. If you choose to modify any of these files, be sure to create a backup copy. Your custom changes should be merged back in with the new updated file after the software update is complete

Initial use

When new users are created, the password is set equal to the login name. For example: the user account “guest” will use the password “guest.”

To maintain system security, it is important that the password be changed as soon as practical. To do this, login as the new user and select change password from the workspace menu. Enter a password and record this information for the new user.

Menu appearance

Engineer Menu			
Window Menu	>		
X Manager Menu	>		
Application Menu	>		
System Menu	>		
		System Commands	
		Configuration	>
		Update Software	>
		Add/Remove Users...	
		Backup/Restore...	

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Add users

Use the following procedure to configure user access to the coprocessor.

⇒ **Note:** If, at any point in the following instructions, you have made an error, selecting the `close` target without selecting `Apply` will exit without saving changes.

Step	Action
1	Select System Menu > Add/Remove Users.
2	Select the <code>User</code> target.
3	Select the <code>Add</code> target.
4	Type the user-login name in the <code>Login Name</code> field.
5	Type full user name, office location, work and home phone number into the field provided.
6	Select the appropriate group target from the menu.
7	Select the <code>Apply</code> target to save changes.
8	Repeat Steps 4 through 7 for each user.
9	Select the <code>close</code> target and the window will close.
10	Select the <code>Apply</code> target to save changes.
11	Select the <code>OK</code> target.

Coprocessor Administration Tasks – Create New Coprocessor Users

Step	Action
12	Select the <code>CLOSE</code> target and the window will close.
13	Select the <code>Exit Xsam</code> target.
14	Select the <code>OK</code> target.

7.2 Delete Coprocessor Users

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is not required.

Menu appearance

Engineer Menu	
Window Menu	>
X Manager Menu	>
Application Menu	>
System Menu	>

System Commands	
Configuration	>
Update Software	>
<u>Add/Remove Users...</u>	
Backup/Restore...	

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Delete users

To deny authorized users access to the coprocessor, use the following:

- ⇒ **Note:** If at any point in the following instructions you have made an error, selecting the `close` target without selecting `Apply` will exit without saving changes.

Step	Action
1	Select <code>Add/Remove Users...</code> from the workspace menu.
2	Select the <code>User</code> target.
3	Select each user to be removed by selecting the icon to left of the user-login name. Note: The icon is highlighted to show selection.
4	Select the <code>Remove</code> target.
5	Select the <code>OK</code> target. Note: This step is required only if multiple users have been selected to be removed.
6	Select the <code>Apply</code> target to save changes.
7	Select the <code>OK</code> target.
8	Select the <code>Close</code> target and the window will close.
9	Select the <code>Exit Xsam</code> target in the lower right corner.
10	Select the <code>OK</code> target.

7.3 Modify Coprocessor User Information

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is not required.

Menu appearance

Engineer Menu	
Window Menu	>
X Manager Menu	>
Application Menu	>
System Menu	>

System Commands	
Configuration	>
Update Software	>
Add/Remove Users...	
Backup/Restore...	

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Modify user information

To modify a user's coprocessor access information, use the following procedure:

- ⇒ **Note:** If, at any point in the following instructions, you have made an error, selecting the `close` target without selecting `Apply` will exit without saving changes.

Step	Action
1	Select <code>Add/Remove Users...</code> from the root menu.
2	Select the <code>User</code> target.
3	Select the user (to modify) by clicking on the icon to the left of the user-login name. Note: icon is highlighted to show selection.
4	If applicable, type changes into the <code>Login ID</code> field.
5	If applicable, type changes into the full name field.
6	To change the access level, select a group target.
7	Repeat Steps 4 through 7 for each user to be modified.
8	Select the <code>Apply</code> target to save changes.
9	Select the <code>Close</code> target and the window will close.
10	Select the <code>Exit Xsam</code> target and the window will close.
11	Select the <code>OK</code> target.

7.4 Backup the Local Coprocessor Hard Drive

Two options

There are several ways to backup and restore the data on the hard drive. Two options will be discussed here. One option is to use the Backup/Restore menu selection. The other option is to enter a command from a terminal connected to the WSI2 I/O board. Both perform the same type of backup (fbbackup).

However, the “menu option” allows other functions to be performed on the Universal Station^X while the backup is running (e.g., someone can be using the Native LCN Window to control the process).

The “terminal option” does not allow other functions to be performed on the Universal Station^X while the backup is running. The terminal option, on the other hand, runs in “single-user mode,” which allows a more error free backup. The menu option usually has a few block checksum mismatch errors. A backup with these errors can still be used to restore the files on the hard drive, but there will be a few files that cannot be restored because of the errors. The terminal option is the preferred method if the station can be “down” during the backup. The backup procedure takes about 45-60 minutes. You must weigh the options and decide which is the better option for your situation.

Required access level

The menu option procedure must be performed while logged in as “engineer”, and does not require the coprocessor (root) password. The terminal option does not require (or allow) login from the login banner, but does require the coprocessor (root) password.

Menu appearance

Engineer Menu	
Window Menu	>
X Manager Menu	>
Application Menu	>
<u>S</u> ystem Menu	>
System Commands	
Configuration	>
Update Software	>
Add/Remove Users...	
<u>B</u> ackup/ <u>R</u> estore...	

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Menu option

Use the following procedure to backup the local coprocessor hard drive using the menu option.

⇒ **Note:** This procedure currently performs only full backups. The contents of the DAT tape will be overwritten.

Step	Action
1	<u>To perform a backup of the local coprocessor hard drive using a local DAT drive.</u> - For local initial configuration, enter the following from the system terminal: <pre>vi /usr/lib/X11/system.Xdefaults</pre> - Change line 232 from <pre>! *XSam*localHost.labelString: <host name></pre> to <pre>*XSam*localHost.labelString:UxS1</pre> Note: UxS1 is the actual hostname of your station. There is no space after the command line. - Reboot the UxS system
2	<u>To perform a backup of the local coprocessor hard drive using a remote DAT drive.</u> - For local initial configuration: - On the local station (the first station without a DAT drive) enter the following from the system terminal: <pre>vi /usr/lib/X11/system.Xdefaults</pre> - Change line 233, 234 or 235 from <pre>! *Xsam*remoteHost2.labelString: <host name></pre> to <pre>*Xsam*remoteHost2.labelString:UxS2</pre> Note: UxS2 is the actual hostname of remote station. There is no space after the command line. - For remote initial configuration: - Telnet to the remote station (the second station with a DAT drive) - Enter the following from the system terminal: <pre>vi /etc/hosts.equiv</pre> - Enter the full name of the local station (the first station without a DAT drive - alias names can not be used). For example, UxS1.honeywell.com - Reboot both stations
3	Select System Menu->Backup/Restore from the workspace menu.
4	Select the Backup target.

Coprocessor Administration Tasks – Backup the Local Coprocessor Hard Drive

Step	Action
5	<u>To perform a backup of the entire Universal Station^X filesystem</u> Select the <code>System(all files)</code> target Note: Generally, this is a better choice than only backing up user files. <u>To perform a backup of Universal Station^X user files only</u> Select the <code>User files only</code> target.
6	<u>To perform a FULL backup of all files within the scope</u> Select the <code>FULL (all files)</code> target. Note: Performing an incremental backup is a feature that is not available at this time.
7	Select the appropriate <code>host</code> target. This indicates which device will have the tape drive.
8	Select the <code>Start Backup</code> target. Note: Depending on your selection (system or user backup), the backup operation will take 2 to 60 minutes to complete.
9	Upon completion, the following message is displayed: <code>fbackup: runtime: 2038 Seconds</code> <code>End of Execution</code>
10	Select the <code>Close</code> target.
11	Select the <code>Exit Xsam</code> target. Click on <code>OK</code> at the message <code>Do you really want to exit?</code>
12	Remove the backup tape from the DAT drive.

Terminal option

Use the following procedure to backup the local coprocessor hard drive using the terminal option.

- ⇒ **Note:** This procedure, as written, performs full backups. The contents of the DAT tape will be overwritten. If you wish to perform incremental backups, you will need to become more familiar with the fbackup command. See the HP-UX document on CD-ROM to learn more about fbackup.

Step	Action
1	Gently Insert the tape into the DAT drive. If you are unable to successfully insert the tape, gently remove the cartridge and reinsert it again. If you are still unsuccessful, contact your Honeywell service representative for assistance.
2	Connect a vt100 terminal to the WSI2 I/O board of the U ^X S.
3	Press <RETURN> (to get a login prompt), and login as root
4	Put the U ^X S in single user mode by entering <code>shutdown -y 0</code> (zero) and pressing <RETURN> . Note: This will take about 30 seconds.
5	Enter <code>/etc/fbackup -i / -f /dev/rmt/0m</code> (zero-em) and press <RETURN> . Note: This command performs a backup of the entire U ^X S filesystem; it is the recommended command. However, if you prefer to backup only a portion of the system, you can specify a directory to be backed up following the “-i” option. Example: <code>/etc/fbackup -i /users/XUS -f /dev/rmt/0m</code> This will backup only the files under the /users/XUS directory (and all of it's subdirectories). This is equivalent to selecting a User backup when using the Menu Option. Note: Depending on the directories you are backing up (/ or /users/XUS), the backup operation will take 2 to 60 minutes to complete.
6	After the backup completes, remove the backup tape from the DAT drive.
7	Enter <code>reboot -q</code> and press <RETURN> .
8	Disconnect the vt100 terminal from the WSI2 I/O board of the U ^X S.

7.5 Restore Local Coprocessor Hard Drive Data

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is not required.

Menu appearance

Engineer Menu	
Window Menu	>
X Manager Menu	>
Application Menu	>
System Menu	>

System Commands	
Configuration	>
Update Software	>
Add/Remove Users...	
<u>B</u> ackup/ <u>R</u> estore...	

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Coprocessor restore menu

This menu provides the mechanism necessary to restore the coprocessor filesystem from a prior backup tape. It allows the user to specify the process by which files are restored.

Restore types

The restore procedure allows two types of file restoration:

- Overwrite — restores tape contents to the Universal Station^X regardless of time stamp.
- Update — writes files with dates later than those present in the Universal Station^X.

Perform a restore using the workspace menu

To perform a restore, follow this procedure:

- ⇒ **Note:** If, at any point in the following instructions, you have made an error, select the EXIT RESTORE target to exit.

Step	Action
1	<u>To restore the local coprocessor hard drive using a local DAT drive.</u> - For local initial configuration, enter the following from the system terminal: <pre>vi /usr/lib/X11/system.Xdefaults</pre> - Change line 232 from <pre>! *XSam*localHost.labelString: <host name></pre> to <pre>*XSam*localHost.labelString:UxS1</pre> Note: UxS1 is the actual hostname of your station. There is no space after the command line. - Reboot the UxS system
2	<u>To restore the local coprocessor hard drive using a remote DAT drive.</u> - For local initial configuration: - On the local station (the first station without a DAT drive) enter the following from the system terminal: <pre>vi /usr/lib/X11/system.Xdefaults</pre> - Change line 233, 234 or 235 from <pre>! *Xsam*remoteHost2.labelString: <host name></pre> to <pre>*Xsam*remoteHost2.labelString:UxS2</pre> Note: UxS2 is the actual hostname of remote station. There is no any space after the command line. - For remote initial configuration: - Telnet to remote station (the second station with a DAT drive) - Enter the following from the system terminal: <pre>vi /etc/hosts.equiv</pre> - Enter the full name of the local station (the first station without a DAT drive - alias names can not be used. For example, UxS1.honeywell.com - Reboot both stations

Coprocessor Administration Tasks – Restore Local Coprocessor Hard Drive Data

Step	Action
3	Remove the appropriate tape from its protective case and gently insert it into the tape drive. Press gently on the edge of the tape cartridge while inserting it into the unit until you feel the unit pull on the tape. The unit will then load and position the tape. While positioning is being performed, both indicator lamps on the front of the unit will flash. When the right lamp shows a solid green, positioning is complete and the tape unit is ready to use. If you are unable to successfully insert the tape, gently remove the cartridge and reinsert it again. If you are still unsuccessful, contact your Honeywell service representative for assistance.
4	Select Backup/Restore... from the workspace menu.
5	Select the Restore target.
6	<u>To perform an overwrite with all files on the tape</u> Select the Overwrite target. WARNING - This procedure may cause error messages to be output during the restore due to conflict with running system processes. It is recommended that selective restores be performed instead of complete restores whenever feasible. <u>To perform an update with files from the tape</u> Select the Update target.
7	Select the Start Restore target.
8	Select the Close target.
9	Select the Exit Xsam target. Click on OK at the message Do you really want to exit?

Perform a restore using SAM

Step	Action
1	Insert the appropriate tape into the local DAT tape drive.
2	Select System Menu > Configuration > System Configuration from the workspace menu.
3	Type the coprocessor (root) password and click OK at the prompt. The System Administration Manager (SAM) window appears
4	Click on Backup and Recovery and select Actions > Open Item .
5	Click on Interactive Backup and Recovery and select Actions > Open Item .
6	Select Actions > Recover Files or Directories .
7	- Click on each of the following buttons within the Recover Files or Directories window and enter the appropriate information for each option: - Specify Tape Device Options ... - Select Recovery Scope ... - Set Additional Parameters ... - Click OK when finished.
8	Select File > Exit .
9	Select File > Exit SAM .

7.6 Backup a Remote Coprocessor Hard Drive

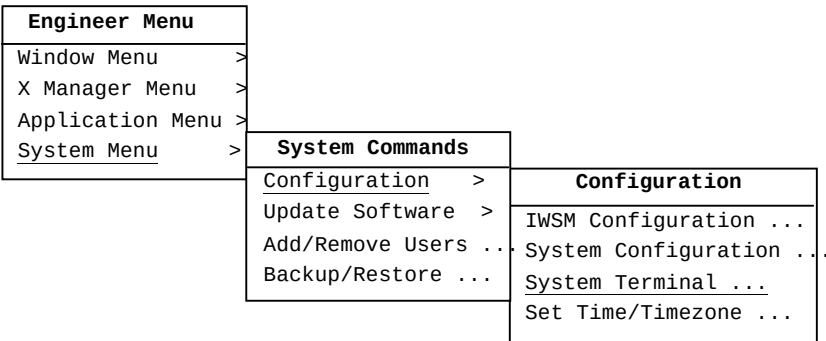
Why this procedure?

This procedure will backup the hard drive that is on a remote coprocessor. That coprocessor can be another Universal Station^X, an Application Module^X, or a Hewlett Packard workstation. The remote device can also be any other device that has the HP-UX 10.20 operating system. You would use this procedure if the remote device is an Application Module^X, or a Hewlett Packard workstation that does not have a DAT drive. If the remote device is a Universal Station^X, either this procedure or the Backup the Local Coprocessor Hard Drive procedure could be used. In the case of the Universal Station^X, the only difference is whether you will be performing the procedure from the Universal Station^X with the hard drive (the other procedure), or from the Universal Station^X with the DAT drive (this procedure). The backup procedure takes about 45-60 minutes.

Required access level

This procedure must be performed while logged in as “engineer”, and requires the coprocessor (root) password.

Menu appearance



51140

Backup procedure

Use the following procedure to backup a remote coprocessor hard drive to the DAT drive on the local Universal Station^X.

⇒ **Note:** This procedure, as written, performs full backups. The contents of the DAT tape will be overwritten. If you wish to perform incremental backups, you will need to become more familiar with the `fbackup` command. Use the documentation distributed with the Universal Station^X on CD-ROM to learn more about `fbackup`.

Step	Action
1	Login as engineer to the U ^X S with the DAT drive.
2	Gently Insert the tape into the DAT drive. If you are unable to successfully insert the tape, gently remove the cartridge and reinsert it again. If you are still unsuccessful, contact your Honeywell service representative for assistance.
3	Select System Menu > Configuration > System Terminal from the workspace menu.
4	Enter the coprocessor (root) password for the local U ^X S and press <RETURN>.
5	Verify that the name of the remote node is in the <code>/.rhosts</code> file of the local station . If it is, go to the next step, if not, follow these instructions to add it: a. Enter <code>vi /.rhosts</code> and press <RETURN>. b. Enter <Shift>g<Shift>a (GA) and press <RETURN>. c. Enter the <u>full</u> name of the remote node (alias names can not be used). d. Press <ESC>. e. Enter <code>:wq!</code> and press <RETURN>.
6	Enter <code>rlogin remotehostname</code> and press <RETURN>. Note: remotehostname is the name of the device with the hard drive.
7	If necessary, enter the root password for the remote device and press <RETURN>.
8	If prompted for TERM (terminal type), enter <code>hpterm</code> and press <RETURN>.

Coprocessor Administration Tasks – Backup a Remote Coprocessor Hard Drive

Step	Action
9	Enter <code>/etc/fbackup -i / -f localhostname:/dev/rmt/0m</code> (zero-em) and press <RETURN>. Note: <i>localhostname</i> must be a host that is known to the remote system. Note: This command performs a backup of the entire U^XS filesystem; it is the recommended command. If you prefer to backup only a portion of the system, however, you can specify a directory to be backed up following the “-i” option. Example: <code>/etc/fbackup -i /users/XUS -f uxs1:/dev/rmt/0m</code> This will backup only the files under the /users/XUS directory (and all of it's subdirectories). Note: Depending on the directories you are backing up (/ or /users/XUS), and the size of your hard drive, the backup operation will take 2 to 60 minutes to complete.
10	After the backup completes, enter <code>exit</code> and press <RETURN>. This will log out of the remote node.
11	Remove the backup tape from the tape drive
12	Log out of the U^XS.

7.7 Restore a Remote Coprocessor Hard Drive

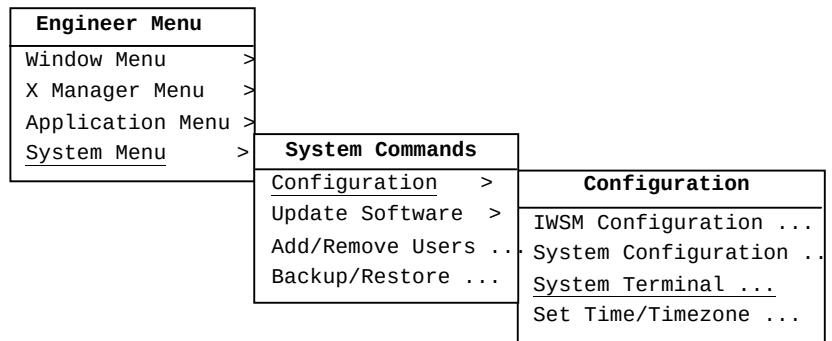
Why this procedure?

This procedure will restore the hard drive that is on a remote coprocessor. That coprocessor can be another Universal Station^X, an Application Module^X, or a Hewlett Packard workstation. The remote device can also be any other device that has the HP-UX 10.20 operating system. You would use this procedure if the remote device is an Application Module^X, or a Hewlett Packard workstation that does not have a DAT drive. If the remote device is a Universal Station^X, either this procedure or the Restore the Local Coprocessor Hard Drive procedure could be used. In the case of the Universal Station^X, the only difference is whether you will be performing the procedure from the Universal Station^X with the hard drive (the other procedure), or from the Universal Station^X with the DAT drive (this procedure). The Restore procedure takes about 45-60 minutes.

Required access level

This procedure must be performed while logged in as “engineer”, and requires the coprocessor (root) password.

Menu appearance



51140

Restore procedure

Use the following procedure to Restore a remote coprocessor hard drive from the DAT drive on the local Universal Station^X.

⇒ **Note:** This procedure, as written, restores all files on the DAT tape. The contents of the hard drive will be overwritten even if the files on the DAT are older than the files on the hard drive. If you wish to perform selective file restores, you will need to become more familiar with the frecover command. See the HP-UX documentation on CD-ROM to learn more about the frecover command.

Step	Action
1	Login as engineer to the U ^X S with the DAT drive.
2	Gently Insert the tape into the DAT drive. If you are unable to successfully insert the tape, gently remove the cartridge and reinsert it again. If you are still unsuccessful, contact your Honeywell service representative for assistance.
3	Select System Menu > Configuration > System Terminal from the workspace menu.
4	Verify that the name of the remote node is in the <code>/.rhosts</code> file of the local station . If it is, go to the next step, if not, follow these instructions to add it: a. Enter <code>vi /.rhosts</code> and press <code><RETURN></code> . b. Enter <code><Shift>g<Shift>a</code> (GA) and press <code><RETURN></code> . c. Enter the <u>full</u> name of the remote node (alias names can not be used). d. Press <code><ESC></code> . e. Enter <code>:wq!</code> and press <code><RETURN></code> .
5	Enter the coprocessor (root) password for the local U ^X S and press <code><RETURN></code> .
6	Enter <code>rlogin remotehostname</code> and press <code><RETURN></code> . Note: remotehostname is the name of the device with the hard drive.
7	If necessary, enter the root password for the remote device and press <code><RETURN></code> .
8	If prompted for TERM (terminal type), enter <code>hpterm</code> and press <code><RETURN></code> .

Coprocessor Administration Tasks – Restore a Remote Coprocessor Hard Drive

Step	Action
9	Enter: <code>/etc/frecovery -rov -f localhostname:/dev/rmt/0m</code> (zero-em) and press <RETURN>. Note: <i>localhostname</i> must be a host that is known to the remote system. Note: This command performs a restore of all the files that are on the DAT tape, and will overwrite newer files on the hard drive with older ones from the DAT tape. If you prefer to Restore only a portion of the files from the tape, you can specify a “graph file” (which indicates which files to be restored) following a “-g” option. Example: <code>frecovery -x -g /tmp/graphfile -f uxs2:/dev/rmt/0m</code> Examples of entries in graphfile are as follows: i /users/XUS e /users/XUS/operator i /usr/lib/X11 i /etc/passwd This will restore (include) all the files under the /users/XUS (except those in the /users/XUS/operator directory). It will also restore (include) all the files under the /usr/lib/X11 directory (and all it's subdirectories), and will restore the /etc/passwd file Note: Depending on the directories you are restoring, and the size of your hard drive, the Restore operation will take 2 to 60 minutes to complete.
10	After the Restore completes, enter <code>exit</code> and press <RETURN>. This will log out of the remote node.
11	Remove the backup tape from the tape drive.
12	Log out of the U ^X S.

7.8 Local Non-Operating System Software Upgrade

Overview of local procedure

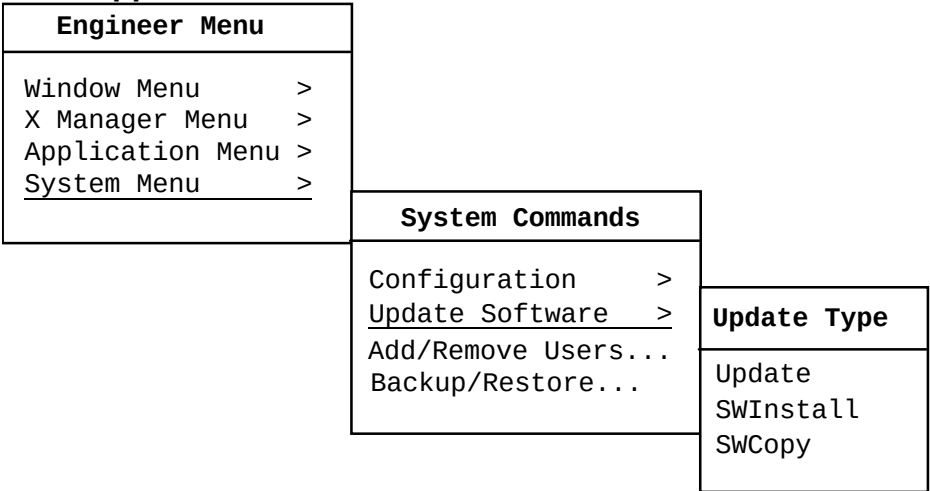
The procedures in this section may be used to upgrade or install non-operating system software on a U^XS node that has a locally connected DAT tape drive. It is assumed that the node on which the software will be installed has already been upgraded to HP-UX 10.20. Upgrading the operating system to HP-UX 10.20 is covered in the *Universal Station^X Customer Release Guide* for Release 300 (UX04-493).

This section provides instructions for installing software using the following two methods:

- SWInstall: used for all Honeywell supplied software
- Update: Used for software supplied by third parties

The installation of software using either of these methods is performed from a nested menu accessed through the System Menu within the Engineer workspace.

Menu appearance



Required configuration

This procedure requires a DAT tape unit connected to the Universal Station^X. The station must be operational with full user-login.

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is required.

Backing up appropriate files

The .baseUser directory in each user group (engineer, operator, etc.) contains the default files that will be used to initially configure a new user’s own directory. Be aware that Honeywell may update these files during a software release upgrade procedure. If you choose to modify any of these files, be sure to create a backup copy. Your custom changes should be merged back in with the new updated file after the software update is complete.

Using SWInstall to upgrade software locally

Step	Action
1	Insert the appropriate software distribution tape into the DAT tape drive.
2	Log in as engineer on the U ^X S that you will be installing the software on.
3	On the U ^X S display, select System Menu > Update Software > SWInstall
4	In the Local Software Install window, type the password for root and click OK.
5	A Note (hostname) window may appear indicating that the default source is not valid because you have not yet specified the source. Click OK to continue.
6	A Specify Source (hostname) window appears. Enter the information for the following options: • Source Depot Type . . . – Click to display additional options – Choose Local Tape • Source Host Name . . . • Source Depot Path . . . • Software Filter: none • Click OK.
7	The SD Install - Software Selection (hostname) window appears with a list of products available on the software distribution tape. • Mark the software for installation by clicking on the software product name. Use Ctrl click to select more than one item. Note: Software products are listed in alphabetical order. • Select Actions > Install (analysis)

Coprocessor Administration Tasks – Local Non-Operating System Software Upgrade

Step	Action
8	An Error (hostname) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies – it is just a warning that dependencies do exist. Click OK to continue.
9	An Install Analysis (hostname) window appears with the following information displayed: • Target: <i>source host name</i> • Status: <i>Analyzing or Ready</i> • Products Scheduled: <i>software products marked for install</i> Press OK when the Status indicates Ready or Ready with Errors.
10	A Confirmation (hostname) window appears indicating that installation will now begin and that only those products which passed analysis will be installed. Click Yes to continue.
11	An Install Window (hostname) window appears with the following information displayed: • Target: • Status: • Percent Complete: • Kbytes Installed: • Time Left (minutes): • Loading Software: You may look at the Logfile when the status indicates Completed to determine if there were any errors during installation. To do so, perform the following: • Click on the Logfile button. A Logfile (hostname) window appears. • Scroll through the log to determine if there were any errors during the installation. • When finished, click OK. • Click Done from the Install window.
12	Select File > Exit to exit SD Install .
13	Remove the software distribution tape from the local tape drive.
14	If your system requires a reboot, perform the following: • Select System Menu > Configuration > System Terminal. • Enter the password for root in the System Terminal window and click OK. • Type the following at the prompt: /etc/reboot

Using Update to upgrade other software

Use the update command for software that is in the update format, if necessary.

Step	Action
1	Insert the appropriate update format tape into the DAT tape drive.

Coprocessor Administration Tasks – Local Non-Operating System Software Upgrade

Step	Action
2	Log in as engineer on the U ^X S that you will be installing the software on.
3	On the U ^X S display, select System Menu > Update Software > Update
4	In the Local Software Install window, type the password for root and click OK. A Local Software Update window appears.
5	• Verify that the Source is Tape Device: /dev/rmt/0m. • If you need to change the software source perform the following: – Use the arrow key to highlight Change Source or Destination -> and then press Enter – From the Change Source or Destination window, highlight From Tape Device to Local System and then press Enter. – Enter the correct Tape Device File (source): – Verify that the Destination Directory is correct. • Click the Done button located on the bottom of the Local Software Update window when finished.
6	• From the Local Software Update window, use the arrow key to highlight one of the following options: – Select All Filesets on the Source Media -> – Select/View Partitions and Filesets • Press Enter.
7	• Use the arrow key to highlight one of the following options: – Start Loading Now . . . – Modify/View Partitions and Filesets – Return to Main Menu • Press Enter.
8	• A screen appears describing the filesets that will be loaded and telling you how to view the logfile. • You will also receive a warning stating that this is your last chance to change your selections before being prompted to: Start loading filesets now? (y or n) • Type y when ready and press Enter.
9	Once the software has been loaded, select File > Exit to exit Update .
10	Remove the software update tape from the local tape drive.
11	If your system requires a reboot, perform the following: • Select System Menu > Configuration > System Terminal. • Enter the password for root in the System Terminal window and click OK. • Type the following at the prompt: /etc/reboot

7.9 Software Distribution Depot Preparation

Overview of procedure

This procedure is used to prepare a Universal Station^X system to act as a Software Distribution Depot. Upgrading non-operating system software on additional remote U^XS nodes can then be accomplished using the depot as the software source for the remote nodes.

You will use the SWCopy utility that is accessed through a nested menu within the Engineer Workspace Menu. The `swcopy` command copies the software from the source (local tape in this procedure) to a depot which can then be used as the installation source for remote nodes on the same network. Software that is copied into a depot cannot be used; it is placed there only to act as a source for other installations.

Menu appearance

Engineer Menu			
Window Menu	>		
X Manager Menu	>		
Application Menu	>		
<u>S</u> ystem Menu	>		
		System Commands	
		Configuration	>
		<u>U</u> ppdate Software	>
		Add/Remove Users...	
		Backup/Restore...	
		Update Type	
		Update	
		SWInstall	
		<u>S</u> WCopy	

Required configuration

This procedure requires a DAT tape unit connected to the Universal Station^X. The station must be operational with full user-login capabilities and connected to an Ethernet network accessible to all Universal Station^Xs to be updated. It is assumed that the node on which the software will be copied and eventually installed has already been upgraded to HP-UX 10.20 (see the *U^XS Customer Release Guide*, UX04-493).

ATTENTION

Universal Station^X stations being used as full software distribution servers must have at least a 1 gigabyte of hard disk drive.

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is required.

Using SWCopy to configure a Software Distribution Depot

Step	Action
1	Log in as engineer to the U ^X S node that will act as the Software Distribution Depot.
2	Insert the appropriate software distribution tape into the local DAT tape drive.
3	On the UXS display, select System Menu > Update Software > SWCopy
4	In the Create Software Depot window, enter the password for root and click OK.
5	A Select Target Depot Path (hostname) window appears. Enter the following information: - Target Depot Path: <code>/var/spool/sw/software_directory</code> where <code>software_directory</code> is the directory name you specify - Click OK
6	Click OK if a note window appears with the following message: "The default source cannot be used as a source because it is already marked as a target."
7	- A Specify Source (hostname) window appears. Enter the following information: - Source Depot Type: Local Tape - Source Host Name . . . <code>hostname</code> - Source Depot Path: <code>/dev/rmt/0m</code> - Software filter: none - Click OK
8	In the SD Copy - Software Selection (hostname) window, perform the following: - Select Options > Change Options. - A Note (hostname) window appears that recommends using caution when changing options. Click OK to continue.

Step	Action
	• Uncheck the Enforce dependency analysis errors in agent. Click OK
9	In the SD Copy - Software Selection (<i>hostname</i>) window perform the following: • Highlight the appropriate software selection line. • Select Actions > Copy (Analysis)
10	An Error (<i>hostname</i>) window may appear indicating an error because pre-requisite filesets are required. Press ENTER to continue
11	Click OK when the Copy Analysis (<i>hostname</i>) window shows a status of Ready with Warnings
12	In the Confirmation (<i>hostname</i>) window, click Yes.
13	A Copy Window (<i>hostname</i>) window appears indicating the following information: • Target: • Status: • Percent Complete: • Kbytes copied: • Time left (minutes): • Copying software: Click Done when the Status indicates completed.
14	Exit SWCopy by selecting File > Exit
15	Remove the software distribution tape from the local tape drive.
16	The software is now loaded on the Software Distribution Depot. When you are ready to load the software onto a remote node, go to the next section, Loading the UXS Personality from a Software Distribution Depot to a Remote Node.

7.10 Remote Non-Operating System Software Upgrade

Overview

In this procedure you will install the non-operating system software on a remote node using HP-UX's *swinstall* command and the Software Distribution Depot configured in the previous procedure.

This procedure is performed through the System Terminal window at the display console of a U^XS that has been configured as the Software Distribution Depot. You will remote login to the client node and then install the software from the Software Distribution Depot. It is assumed that the node to which the software will be installed is on line and has already been upgraded to HP-UX 10.20.

Procedure requirements

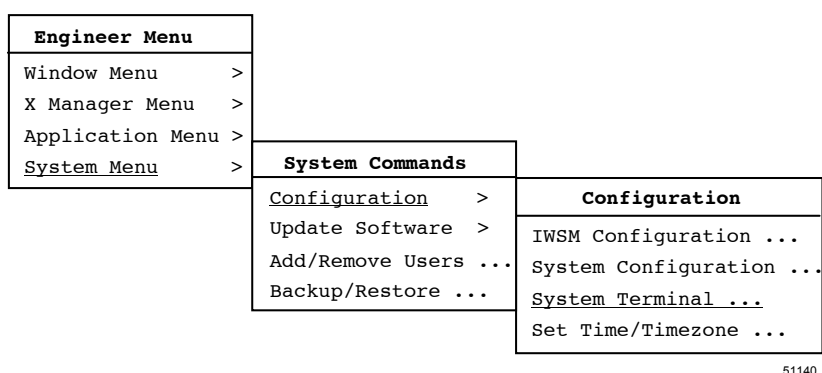
This procedure requires that an additional Universal Station^X be connected to the LAN. This software distribution depot must already have been prepared using the procedure in Section 7.9, Software Distribution Depot Preparation.

Before beginning this procedure, verify that you have all the necessary preliminary information for both the client and the host.

Required access level

The procedures must be performed while logged in as “engineer.” The coprocessor (root) password is required.

Menu appearance



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Using SWInstall to upgrade software remotely

Step	Action
1	Login as engineer to the U ^X S node that contains the Software Distribution Depot.
2	On the UXS display, select System Menu > Configuration > System Terminal
3	In the System Terminal window, enter the password for root and click OK if you are not already logged in.
4	Type the following at the prompt: <i>rlogin client_node_name</i> Where <i>client_node_name</i> is the actual name of the node that will have the software installed.
5	Enter the password of the client node and press ENTER.
6	In the remote System Terminal window, enter the following command: <code>/usr/sbin/swinstall</code>
7	A screen of instructions explaining how to use <code>swinstall</code> through the terminal appears. Press ENTER to continue.
8	The SD Install - Software Selection (hostname) window appears within the

Step	Action
	System Terminal window with a note stating that the software source does not exist. You will define the software source in the next step. Press ENTER to continue.
9	A Specify Source (hostname) window appears with the entry in the Source Depot Type highlighted. Select the appropriate software source and path by performing the following: • Press Enter to display additional options, if necessary. • Use the Up/Down Arrow keys to highlight your software source of Network Directory and press ENTER. • Use the TAB key to go to Source Depot Path. Enter the Target Depot Path defined in the previous procedure. • Press the TAB key to go to OK and then press ENTER.
10	You will return to the SD Install - Software Selection window. Mark the software for installation by performing the following: • Highlight the software product by pressing the space bar. • Press the TAB key to go to the menu bar. • Use the right arrow key to move to the Actions drop down menu and then press ENTER. • Use the down arrow key to select Install (Analysis) from the Actions menu and then press ENTER.
11	An Error (hostname) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies – it is just a warning that dependencies do exist. Press ENTER to continue
12	An Install Analysis (hostname) window appears with the following information displayed: • Target • Status • Products Scheduled The install program is checking for adequate disk space and software dependencies. If all dependencies have been met and there is adequate disk space, the Status will indicate Ready. • If the Status does not indicate Ready, you will need to correct the problem before continuing. • Press the TAB key to go to OK and then press ENTER.
13	A Confirmation (hostname) window appears with the installation information displayed. Press ENTER to continue.

Step	Action
14	An Install Window (hostname) window appears during the installation process with the following information displayed: • Target • Status • Percent Complete • Kbytes Installed • Time Left (minutes) • Loading Software information The Percent Complete will advance to 100%. Do not, however, begin the next step until the Status indicates Completed or Ready.
15	At this time you can choose to look at the logfile to determine if there were any errors during installation. To do so, perform the following: • Press the TAB key to go to Logfile • Press ENTER to display all the loaded modules. • A Logfile (hostname) window appears. • Scroll through the log to determine if there were any errors during the installation. When finished looking at the logfile, press the TAB key to go to OK and then press ENTER.
16	You will be returned to the Install (hostname) window. Press the TAB key to go to DONE and then press ENTER.
17	Exit <code>swinstall</code> by pressing the TAB key to go to the command bar and selecting File > Exit

7.11 Checking Tracking Files

Software tracking

There are several utilities within UNIX which provide troubleshooting information by constantly providing on-line recording of certain events/messages. This information can often help in locating the trouble source.

Interpreting the file contents is beyond the scope of this publication; however, the location and use of these files will be described here.

Login logs

The system logs all login attempts in the `/etc` directory. You may check the logs to determine if there have been any attempts to gain unauthorized access.

Login attempt	Recorded in log file
Successful Login	<code>/etc/wtmp</code>
Failed Login	<code>/etc/btmp</code>

The files are stored in binary format. Use the UNIX command `last` to read the `wtmp` file (successful logins and logouts) and the command `lastb` to read the `btmp` file (failed logins).

For example,

```
last wtmp <RETURN>
lastb btmp <RETURN>
```

Shutdown log

The system logs all shutdowns in the file `/usr/adm/shutdownlog`. Check this file for any unattended shutdowns (power outage) or any abnormal conditions incurred during past shutdowns.

7.12 Software Restoration from Local DAT

Overview

This procedure will restore the software from DAT tapes to the U^XS's hard drive. It can be used when the software on the hard disk has been corrupted, or when the hard disk has been replaced with one that has no software. This procedure requires the U^XS to have a DAT drive. If it does not, use the "Software Restoration from a Server" procedure. The procedure in this section is for restoring the HP-UX 10.20 operating system and other Release 300 software if you do not have adequate backup and restore tapes.

This procedure should take about 1.5 to 2.5 hours.

Required

In order to complete this procedure, you must have:

- vt100 terminal (or a terminal that emulates vt100)
- cable that connects the vt100 to the WSI2 I/O board
- DAT tapes:
 - HP-UX Install (U^XS R300)
 - U^XS Windows SW
 - U^XS Personality for U^XS R300
- Hostname and IP address of the destination U^XS
- Hostname and IP address of the server U^XS
- Coprocessor (root) password of the server U^XS

Preferred (in addition to required)

- DAT tapes:
 - System backup
 - Additional software distribution tapes

Optional (in addition to required)

In order to complete this procedure, you might want to have:

- Subnet mask for your LAN
- Hostname and IP address of the gateway
- Hostname and IP address of the BIND server

Restore operating system procedure

Step	Action
1	Connect the console terminal and DAT drive to the U ^X S (see the U ^X S Service Manual).
2	Insert the U ^X S Install tape for HP-UX 10.20 into the DAT tape drive.
3	Log in to the U ^X S as root through the local console terminal and enter the root password.
4	Verify that you have all of the system configuration settings for the U ^X S.
5	Type the following at the prompt: /etc/reboot
6	Press the ESC key immediately when you see the following message: Press [ESC] to discontinue the boot process
7	The following menu appears: <pre> Auto Boot aborted. ==== MAIN MENU ===== Key Operation ----- 1 Boot From a Device 2 Path Configuration 3 Mode Configuration 4 Interactive Testing 5 Firmware Information 6 Hardware Information 7 System Configuration ... 77 Reset the System 88 Change Mode 99 Restart Auto Boot ----- Press Key, then press [Enter/Return] Select Boot From a Device by pressing 1 and then ENTER </pre>
8	• Select the appropriate tape device: – 2 if single disk system – 3 if dual disk system • Press ENTER

Step	Action
9	A Welcome to Ignite-UX window appears. This Welcome screen is the first of several screens that will guide you through the installation process. • Press the TAB key to highlight items on the screen and then press ENTER to choose or activate the highlighted item. • Use the Up or Down Arrow keys to navigate through the items in the menu.
10	Select the Install HP-UX option and press ENTER.
11	The User Interface and Media Options window appears. • The default for Source Location Options is Media Only Installation • The default for User Interface Options is Guided Installation • Choose these defaults by pressing the TAB key to go to OK and pressing ENTER The system will load the configuration user interface.
12	The Install HP-UX wizard: Select an overall system configuration window appears. • The default Configuration is HP-UX B.10.20 Default • Accept this default by pressing the TAB key to go to NEXT and then pressing ENTER
13	The Install HP-UX wizard: Select a system environment window appears. • Press the space bar to see additional options • Select the Archive UXS 30 with HP-UX environment and press ENTER
14	The Install HP-UX wizard: Select a root disk window appears. • Select the disk at the appropriate SCSI address • Press the TAB key to go to NEXT and then press ENTER

Step	Action
15	The Install HP-UX wizard: Specify the amount of root swap space window appears. The Install HP-UX wizard determines memory size and calculates a recommended device swap disk size that will be used unless you change it. Honeywell recommends that the root swap space be at least 3 times the size of the physical memory. Perform the following if you wish to change the value calculated by the wizard: • Press ENTER to display a list of predefined selections for swap disk size. • Use the Up/Down Arrow key to navigate through the selections. • Go to OK and press ENTER. • To continue to the next menu, press the TAB key to go to NEXT and then press ENTER.
16	The Install HP-UX wizard: Select a file system type window appears. • Make one of the following sections based upon the size of your hard drive: – If the hard disk is less than 4G then select Whole Disk with HFS as the File System Type. – If the hard disk is 4G or greater then select Logical Volume Manager with HFS as the File System Type. – If the hard disk is at least 2 GB, but less than 4 GB, you may choose either option. • To continue, press the TAB key to go to NEXT and then press ENTER.
17	If you selected Logical Volume in the previous step, the Install HP-UX wizard: Specify root volume group disks window appears. • Select the install program default. • Press the TAB key to go to NEXT and then press ENTER
18	The Install HP-UX wizard: Select a language(s) window appears. • Select the default or display additional options by pressing ENTER • To continue, press the TAB key to go to NEXT and then press ENTER
19	The Install HP-UX wizard: Select Additional Software window appears. • There is no additional software • Press the TAB key to go to NEXT and then press ENTER
20	The Install HP-UX wizard: Pre-install disk information window appears. • Select the install program default • Press the TAB key to go to NEXT and press ENTER
21	The Install HP-UX wizard: Pre-install check information window appears. A new window with WARNING message may appear to let you know that all the data in the disk will be destroyed. • DO NOT continue if you have not performed a full backup of your system. See the Backup Strategy section of this manual. • If you performed a full backup of your disk before starting this installation, press the TAB key to go to NEXT and then press ENTER.
22	The Install HP-UX wizard: System Summary window appears. • Select Show Summary and press ENTER to display the General Summary and Hardware Summary tab. • Press the TAB key to go to the selected window/topic and use the Up/Down Arrow key to scroll through additional information.

Step	Action
	If you are satisfied with your choices , press the TAB key to go to FINISH and then press ENTER. This phase of the installation will take about 30 to 45 minutes. The system will boot at the completion of this step.
23	The initial installation is complete when you see the following message: <pre> ----- - Welcome to HP-UX! Before using your system, you will need to answer a few questions. The first question is whether you plan to use this system on a network. Answer "yes" if you have connected the system to a network and are ready to link with a network. Answer "no" if you: * Plan to set up this system as a stand alone (no networking). * Want to use the system now as a stand alone and connect to a network later. ----- - Are you ready to link this system to a network? Press [y] for yes or [n] for no, then press Return. </pre> See the next step for information on the appropriate response.

Step	Action
24	Note: If you wish to perform remote upgrades, you must configure networking at this time as subsequent procedures will use the configuration information you supply in this step. • Press Y to link this system to a network. • Supply the following information when prompted: – System name and IP address – Root password – Time zone – Any subnets – Subnet mask – DNS server – NIS configuration – Gateway hostname and IP address • If you answer N, you will be prompted for this information only: – System name – Root password – Time zone • Follow the instructions on the screen to configure all necessary information. • Skip configuration of the Font Server if prompted for this information Note: It is possible to change or enter the configuration information at a later time using SAM.
25	At the completion of this section, a congratulatory message appears. The system will automatically reboot and the following message appears: HP-UX Start-up in progress This takes about 2 to 3 minutes. • A column of functions (tasks) and a corresponding status column provides status indicators such as: – OK – N/A – FAIL. • If you configured networking, the status column should display either OK or N/A. • If you did not configure networking, the following items will indicate FAIL: – Configure LAN interfaces – NCS broker daemons – Start audio server daemon
26	Remove the U ^X S Install tape from the tape drive.

Restore U^XS Personality procedure

Step	Action
1	Insert the appropriate U ^X S Personality for UXS R300 tape into the DAT tape drive.
2	Log in to the U ^X S as root through the local console terminal and enter the root password.
3	Type the following at the prompt: /usr/sbin/swinstall
4	The message Starting the terminal version of swinstall... appears with instructions for navigating within the menus using the keyboard.. • If you are not familiar with the use of the keyboard for navigation, read the instructions. • Press ENTER to continue.
5	The install software will scan the default source depot type and will display a Note (hostname) window with a message displayed and with the OK selection highlighted. Press ENTER to continue.
6	A Specify Source (hostname) window appears with the entry in the Source Depot Type highlighted. Select the appropriate software source and path by performing the following: • Press Enter to display additional options, if necessary. • Use the Up/Down Arrow keys to highlight the software source of Local Tape and press ENTER. • Use the TAB key to go to Source Depot Path and enter the correct path, if necessary. • Press the TAB key to go to OK and then press ENTER.
7	The SD Install - Software Selection (hostname) window appears. The UXS- PERSONALITY software should be listed in the window. Mark the software for installation by performing the following: • Highlight the software product by pressing the space bar. • Press the TAB key to go to the menu bar. • Use the right arrow key to move to the Actions drop down menu and then press ENTER. • Use the down arrow key to select Install (Analysis) from the Actions menu and then press ENTER.
8	An Error (hostname) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Press ENTER to continue

Step	Action
9	An Install Analysis (hostname) window appears with the following information displayed: • Target • Status • Products Scheduled The install program is checking for adequate disk space and software dependencies. If all dependencies have been met and there is adequate disk space, the Status will indicate Ready. • If the Status does not indicate Ready, you will need to correct the problem before continuing. • Press the TAB key to go to OK and then press ENTER.
10	A Confirmation (hostname) window appears providing you with installation information. Select Yes and press ENTER.
11	An Install Window (hostname) window appears during the installation process with the following information displayed: • Target: <i>hostname</i> • Status: series of status messages • Percent Complete: advances to 100% • Kbytes Installed: • Time Left (minutes): • Loading Software: software product name The Percent Complete will advance to 100%. Do not, however, begin the next step until the Status indicates Completed or Ready.
12	At this time you can choose to look at the logfile to determine if there were any errors during installation. To do so, perform the following: • Press the TAB key to go to Logfile. • Press ENTER to display all the loaded modules. • A Logfile (hostname) window appears. • Scroll through the log to determine if there were any errors during the installation. • When finished looking at the logfile, press the TAB key to go to OK and then press ENTER.
13	You will be returned to the Install Window (hostname) window. Press the TAB key to go to DONE and then press ENTER.
14	The SD Install - Software Selection (hostname) window appears. • Press the TAB key to go to the File menu and press ENTER. • Use the down arrow key to go to Exit and press ENTER.

Step	Action
15	You must reboot the node for the UXS to be functional. Type the following at the prompt: <code>/etc/reboot</code>
16	When the reboot is complete, the UXS will be functional.
17	• Validate the settings of the <code>/var/adm/inetd.sec</code> file for node security. The file that is provided on the update image does not have any security configured. • There is also a template file available which has maximum security configured, including the prevention of remote logins. Be careful if you use this file: <code>/var/adm/inetd.sec.usx</code>

Restore other software procedure

After the operating system and the U^XS Personality software have been restored to the hard drive, the U^XS is fully functional. Any additional software can be restored locally or remotely using the procedures found in these sections of this manual:

- Section 7.8, "Local Non-Operating System Software Upgrade"
- Section 7.9, "Software Distribution Depot Preparation"
- Section 7.10, "Remote Non-Operating System Software Upgrade"

⇒ **Note:** Section 7.8 contains procedures for loading software using both the SWInstall and the Update utilities.

Alternatively, if you have adequate backup and recovery tapes, you may use the procedures found in these sections of this manual:

- Section 7.5, "Restore Local Coprocessor Hard Drive Data"
- Section 7.7, "Restore a Remote Coprocessor Hard Drive"

Restore completed

At this point, all the software is loaded on the U^XS s hard drive. The U^XS also has minimal configuration. If you did not have a system backup on DAT tape with all your configuration settings, you must “manually” configure your system to recover your database.

Manual data base recovery procedure

Step	Action
1	“Manually” configure your system. Note: You are here because you do not have a DAT tape containing a system backup; therefore, you must configure your system from scratch. Follow the appropriate sections in the U^XS System Administration Manual to complete the configuration. Examples of procedures you must follow: - Configure Coprocessor LAN Information - Configure LAN Routing - Configure Security for Remote Users - Configure Coprocessor Printer - Create New Coprocessor Users - Industrial Workspace Manager - CRT Configuration (for dual CRT U^XS s)
2	If you have a DAT tape containing a user backup, restore it by using the procedure in Section 7.5, "Restore Local Coprocessor Hard Drive Data". If not, “manually” configure user files. This can be done by using <code>ftp</code> to copy user files from other U^XS s (if available), or by following the procedures in Section 6, "Coprocessor User Environment Configuration."
3	Install any optional software by following the instructions that came with that software.
4	Perform a complete backup of your system by following the procedure in Section 7.4, "Backup the Local Coprocessor Hard Drive."

7.13 Software Restoration from a Server

Overview

This section contains the procedures for configuring an upgrade server in order to restore remote nodes. It is intended to be used when the U^XS with the corrupted file system does not have a DAT drive. Another U^XS (which does have a DAT drive) will be required to perform this procedure. The U^XS with the DAT drive will be referred to as the local server U^XS. The U^XS with the corrupted file system will be referred to as the remote client U^XS.

The procedures in this section include the following:

- Installing U^XS Ignite Software on a U^XS station to act as a server
- Configuring the U^XS station to act as an Ignite-UX Server
- Installing HP-UX 10.20 to a remote client node using the Ignite-UX Server

Additional documentation

See also additional information in Ignite-UX Server chapters of the HP documentation at <http://www.docs.hp.com>. To find this information at this site, select *HP-UX Operating System, HP-UX Release 10.x Documentation, Installing and Updating HP-UX 10.x*, and then select the appropriate chapter. For online information about the Ignite-UX server, please see the `/opt/ignite/share/doc/` directory on your system, and the manpage *ignite(5)*.

Required

In order to complete this procedure, you must have:

- U^XS (server) with a 1.2G (or larger) hard drive and a DAT drive
- DAT tapes:
 - U^XS Install Tape for HP-UX 10.20
 - U^XS Core Tape for HP-UX 10.20
- Hostname and IP address of the destination U^XS
- Hostname and IP address of the server U^XS
- Coprocessor (root) password of the server U^XS

Preferred (in addition to required)

In order to complete this procedure, is preferable if you have:

- DAT tapes:
 - System backup tapes

Optional (in addition to Required)

In order to complete this procedure, you might want to have:

- Subnet mask for your LAN
- Hostname and IP address of the gateway
- Hostname and IP address of the BIND server

Requirements

Ignite UX Server Node must have a hard drive of 1 GB or larger

Ignite UX Server Node must already have been upgraded to HP-UX 10.20

Host name and individual IP address of each node to be upgraded

Host name and individual IP Address of the node that will act as the Ignite U^XS Server

ATTENTION

This procedure requires the server U^XS to have a 1 gigabyte (or larger) hard disk drive. Although it is possible to restore from a server U^XS with a 525mb hard drive (small portions each time), this procedure will not discuss that method.

Installing Ignite-UX on a local server node

In this procedure you will install the Ignite-UX Server software, which is on the UXS Core tape, onto a local node using the SWInstall utility. SWInstall can be accessed through a nested menu within the Engineer Workspace Menu. After installing the necessary software and configuring the local UXS node, it can then be used as an Ignite-UX Server in order to restore remote nodes.

This procedure is performed through the display console at a U^XS that has a local DAT tape drive and a hard drive with at least 150 MB of available disk space.

⇒ **Note:** Loading the UXS Install tape will automatically install Ignite-UX.

Step	Action
1	Login as engineer on the U ^X S node that will be used as the Ignite-UX Server.
2	Insert the UXS Core tape into the local DAT tape drive.
3	On the UXS display, select System Menu > Update Software > SWInstall

Step	Action
4	In the Local Software Install window, type the password for root and click OK.
5	A Note (<i>hostname</i>) window may appear indicating that the default source is not valid because you have not yet specified the source. Click OK to continue.
6	A Specify Source (<i>hostname</i>) window appears. Enter the information for the following options: • Source Depot Type . . . – Click to display additional options – Choose Local Tape • Source Host Name . . . <i>hostname of local node</i> • Source Depot Path . . . • Software Filter: none • Click OK.
7	A SD Install - Software Selection (<i>hostname</i>) window appears with a list of products available on the UXS Core tape. • Mark the following software for installation by clicking on the software product name. Use Ctrl click to select more than one item. – Ignite-UX-10-20 – UXS-IGNITE Note: Software products are listed in alphabetical order. • Select Actions > Install (analysis)

Step	Action
8	An Error (hostname) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Click OK to continue
9	An Install Analysis (hostname) window appears with the following information displayed: Target: <i>source host name</i> Status: <i>Analyzing or Ready</i> Products Scheduled: <i>software products marked for install</i> Click OK when the Status indicates Ready or Ready with Errors.
10	A Confirmation (hostname) window appears indicating that installation will now begin and that only those products that passed analysis will be installed. Click Yes to continue.
11	An Install (hostname) window appears with the following information displayed: Target: Status: Percent Complete: Kbytes Installed: Time Left (minutes): Loading Software: You may choose to look at the Logfile when the status indicates Completed to determine if there were any errors during installation. To do so, perform the following: • Click on the Logfile button. A Logfile (hostname) window appears. • Scroll through the log to determine if there were any errors during the installation. • When finished looking at the logfile, click OK. • Click Done from the Install Window.
12	Select File > Exit to exit SD Install.
13	Remove the U^XS Core tape from the local tape drive.

Restore system backup

Step	Action
1	If you do NOT have a DAT tape with a system backup on it, continue with the next procedure (Server U ^X S —procedure 2). If you do, then login as root at the system console.
2	Insert the backup tape in the server U ^X S's DAT drive.
3	At the system console connected to the destination station, type <code>vi /etc/hosts</code> and press <code><RETURN></code> .
4	Enter <code><Shift>ga</code> (GA) and press <code><RETURN></code> .
5	Enter the IP address and host name of the server U ^X S. Ex: 164.145.114.52 trn2 Note: Limit the host name to 8 characters.
6	Press <code><Esc></code>
7	Enter <code>:wq!</code> and press <code><RETURN></code> .
8	Verify that the name of the destination station node is in the <code>/.rhosts</code> file of the server U ^X S'. If it is, go to the next step, if not, follow these instructions to add it: a. At the server U ^X S, select: System Menu->Configuration->System Terminal from the workspace menu. b. Enter the coprocessor (root) password and press <code><RETURN></code> . c. Enter <code>vi /.rhosts</code> and press <code><RETURN></code> . d. Enter <code><Shift>ga</code> (GA) and press <code><RETURN></code> . e. Enter the <u>full</u> name of the destination node (alias names cannot be used). f. Press <code><ESC></code> . g. Enter <code>:wq!</code> and press <code><RETURN></code> . h. Enter <code>exit</code> and press <code><RETURN></code> to close the system terminal window on the server U ^X S.
9	Type <code>/etc/frecovery -rov -f svrhstnm:/dev/rmt/0m</code> and press <code><RETURN></code> . (where <code>svrhstnm</code> is the host name of the server U ^X S). Ex: <code>frecovery -rov -f trn2:/dev/rmt/0m</code> Note1: 0m is zero m, not oh m Note2: This will take about 60-75 minutes.
10	After the frecover has completed, type <code>reboot -q</code> and press <code><RETURN></code> . This completes the procedure—your station has been restored to the state it was in at the time of your last system backup.

Configuring the Ignite-UX Server

In this procedure you will configure the local node on which you previously installed Ignite-UX Server software. Configuring the node will complete the steps necessary to use the node as an Ignite-UX Server in order to restore remote nodes.

This procedure is performed through the System Terminal window at the display console of a U^XS that has the Ignite-UX Server software installed on it.

Step	Action
1	On the U ^X S display, select System Menu > Configuration > System Terminal
2	In the System Terminal window enter the password for root and press OK.
3	You will now need to edit the <code>UxS_300_archive_cfg</code> file in order to set the <code>nfs_source</code> to the IP address of the node acting as the Ignite-UX Server. To do so perform the following from the System Terminal window: - Type the following command: <code>vi /var/opt/ignite/data/Rel_B.10.20/UxS_300_archive_cfg</code> Note: The U and S are uppercase; x is lowercase - Locate the following line identifying the <code>nfs_source</code>: <code>nfs_source = "IP_Address:/var/opt/ignite/archives/UxS.300.0"</code> - Change the <code>IP_Address</code> to the IP address of the node that will act as the Ignite-UX Server. - Save the modified file and exit the editor.
4	From the System Terminal window, type the following command: <code>/usr/sbin/sam</code> The System Administration Manager menu appears.
5	In the SAM window: - Click on Networking and Communications and select Actions > Open Item - Click on Networked File Systems and select Actions > Open Item - Click on Exported Local File Systems and select Actions > Run.
6	From the Networked File System (hostname) window, select Action > Add Exported File System
7	The Add Exported File System (hostname) window appears. You will need to add the local directories to the Exported Local File Systems list by entering the information specified in the following 3 steps.
8	In the Add Exported File System window, make the following selections: - Local directory name: <code>/var/opt/ignite/clients</code> - Choose Option for Unknown User ID: - Specify UID for unknown - User ID: 2 - Accept the defaults for other options - Click Apply
9	In the Add Exported File System window, make the following selections:

Step	Action
	• Local directory name: /var/opt/ignite/data/Rel_B.10.20 • Click Specify User Access button • In the User Access (hostname) window make the following selections: – Allow Access From: All Systems – Access Types: Read-only – Click OK • Click Apply
10	In the Add Exported File System window, make the following selections: • Local directory name: /var/opt/ignite/archives/UxS.300.0 • Click Specify User Access button • In the User Access (hostname) window make the following selections: – Access Types: Read-only – Click OK • Click Apply
11	Click Cancel to return to the Add Exported File System (hostname) window.
12	Select File > Exit to return to the SAM window.
13	Select File > Exit to exit SAM.

Restoring the operating system remotely using an Ignite-UX Server

More detailed information on installing from an Ignite-UX Server is contained in Chapter 5 of the *Installing and Updating HP-UX 10.x HP Documentation*. Use this procedure when you need to restore the operating system on a U^XS node using the Ignite-UX server that has Ignite-UX and UXS-IGNITE installed.

This procedure is performed through the System Terminal window at the display console of a U^XS that contains the Ignite-UX and UXS-Ignite software. Before beginning this procedure, verify that you have all the necessary preliminary information for both the client and the host. This restoration procedure will reformat the hard disk drive and install U^XS release 300 on the node to be upgraded. Any customization done by the end user will have to be repeated after the U^XS release 300 software is restored.

Step	Action
1	Log in as engineer to the U ^X S node that is acting as the Ignite-UX server.
2	On the U ^X S display, select System Menu > Configuration > System Terminal
3	In the System Terminal window, enter the password for root and click OK if you are not already logged in.
4	In the System Terminal window, type the following command: <code>/opt/ignite/bin/bootsys -w -v System_Name</code> (where <code>System_Name</code> is the name of the remote node to be upgraded)
5	Select Y and press ENTER when the following prompt appears: "Would you like bootsys to set the .rhosts file on that system at this time? ([y]/n)"
6	Enter the root password of the remote node to be upgraded. You will receive a series of screen messages.
7	In the System Terminal window, type the following command: <code>/opt/ignite/bin/ignite</code>
8	- Click OK if a window appears with the following message: No clients found A Welcome to Ignite UX screen appears. - Click OK.
9	An Ignite UX (hostname) window appears. The window displays a series of node icons. It may take a few minutes for the icons to appear. Click on the icon that represents the remote node that is to be upgraded.
10	- Select Actions > Install Client > New Install An opt/ignite/bin/itool window appears with a System Hardware Inventory screen. - Click OK. The opt/ignite/bin/itool screen appears with a series of tabs across the top.
11	Select the Basic tab.

Step	Action
12	Make the following selections from within the Basic tab: • Configurations: UxS 300 with HP-UX 10.20 • Root Disk setting and File System Type: – If the disk is 2 GB or less, set the File System Type to Whole Disk with HFS. – If the disk is 4GB or greater, set the File System Type to Logical Volume Manager with HFS. – If the disk is at least 2 GB, but less than 4 GB you may choose the Whole Disk or Logical Volume Manager option. • Validate that the Root Swap size is set to at least three times the size of physical memory • Press Apply
13	Select the System tab.
14	Under Final System Parameters select Set Parameters Now and fill in the information for all the fields.
15	Set the following configurations in the System tab: • Time Zone • Root Password • Network Services
16	Select the DNS tab.
17	• Verify that the gateway and DNS information is properly set. • Click Go!
18	An itool Confirmation (hostname) window appears with a warning message stating that "Continuing the installation will destroy any existing data on this disk." • If the only warning is that the disk appears to contain HP-UX – Press Go! to continue. – Otherwise, correct the problem before continuing
19	• In the Ignite-UX window, select Action > Client Status to bring up a Client Status window where you may monitor the upgrade process. • The left column of each row will display COMPLETE when the upgrade is finished. Click OK when the upgrade is complete. Note: The upgrade will take approximately 20 minutes if there are no slow gateways.
20	The node may contain some warnings as a result of the upgrade. These warnings are OK.
21	From the Ignite-UX window select File > Exit .

Restore U^XS Personality

Before the remote client node is fully functional, you must restore the U^XS Personality software. This procedure is only if you do not have a system backup tape from which to restore your software.

Step	Action
1	Load the U ^X S Personality onto a Software Distribution Depot using the procedure in Section 7.9, "Software Distribution Depot Preparation."
2	Log in as engineer to the U ^X S node that contains the U ^X S Personality software loaded on the Software Distribution Depot if you have not already done so.
3	On the U ^X S display, select System Menu > Configuration > System Terminal
4	In the System Terminal window, enter the password for root and click OK if you are not already logged in.
5	Type the following at the prompt: <pre>rlogin client_node_name</pre> Where <i>client_node_name</i> is the actual name of the node that will have the software installed.
6	Enter the password of the client node and press ENTER.
7	In the remote System Terminal window, enter the following command: <pre>/usr/sbin/swinstall/</pre>
8	A screen of instructions explaining how to use <code>swinstall</code> through the terminal appears. Press ENTER to continue.
9	The SD Install - Software Selection (<i>hostname</i>) window appears within the System Terminal Window with a note stating that the software source does not exist. You will define the software source in the next step. Press ENTER to continue.
10	A Specify Source (<i>hostname</i>) window appears with the entry in the Source Depot Type highlighted. Select the appropriate software source and path by performing the following: • Press Enter to display additional options, if necessary. • Use the Up/Down Arrow keys to highlight the software source of Network Directory and press ENTER. • Use the TAB key to go to Source Depot Path. Enter the Target Depot Path defined in the previous procedure. • Press the TAB key to go to OK and then press ENTER.

Step	Action
11	You will return to the SD Install - Software Selection (<i>hostname</i>) window. The UXS- PERSONALITY software should be listed in the window. Mark the software for installation by performing the following: • Highlight the software product by pressing the space bar. • Press the TAB key to go to the menu bar. • Use the right arrow key to move to the Actions drop down menu and then press ENTER. • Use the down arrow key to select Install (Analysis) from the Actions menu and then press ENTER.
12	An Error (<i>hostname</i>) window may appear if the selected software has any dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Press ENTER to continue
13	An Install Analysis (<i>hostname</i>) window appears with the following information displayed: • Target • Status • Products Scheduled The install program is checking for adequate disk space and software dependencies. If all dependencies have been met and there is adequate disk space, the Status will indicate Ready . • If the Status does not indicate Ready, you will need to correct the problem before continuing. • Press the TAB key to go to OK and then press ENTER.
14	A Confirmation (<i>hostname</i>) window appears with the installation information displayed. Press ENTER to continue.
15	An Install Window (<i>hostname</i>) window appears during the installation process with the following information displayed: • Target: <i>hostname</i> • Status: series of status messages • Percent Complete: advances to 100% • Kbytes Installed: • Time Left (minutes): • Loading Software: software product name The Percent Complete will advance to 100%. Do not, however, begin the next step until the Status indicates Completed or Ready.

Step	Action
16	At this time you can choose to look at the logfile to determine if there were any errors during installation. To do so, perform the following: • Press the TAB key to go to Logfile. • Press ENTER to display all the loaded modules. A Logfile (hostname) window appears. • Scroll through the log to determine if there were any errors during the installation. When finished looking at the logfile, press the TAB key to go to OK and then press ENTER.
17	You will be returned to the Install Window (hostname) window. Press the TAB key to go to DONE and then press ENTER.
18	The SD Install - Software Selection (hostname) window appears. • Use the TAB key to go to the File Menu and press ENTER. • Use the down arrow key to go to Exit and press ENTER.
19	You must reboot the remote node. Type the following at the prompt: <pre>/etc/reboot</pre>

Restore completed

At this point, all the software is loaded on the destination U^XS s hard drive. The destination U^XS also has minimal configuration. Because you do not have a system backup on DAT, you must “manually” configure your system in order to recover your data base. Manual Data Base Recovery Procedure

Step	Action
1	“Manually” configure your system. Note: You are here because you do not have a DAT tape containing a system backup; therefore, you must configure your system from scratch. Follow the appropriate sections in the U^XS System Administration Manual to complete the configuration. Examples of procedures you must follow: – Configure Coprocessor LAN Information – Configure LAN Routing – Configure Security for Remote Users – Configure Coprocessor Printer – Create New Coprocessor Users – Industrial Workspace Manager – CRT Configuration (for dual CRT U^XSs)
2	If you have a DAT tape containing a user backup, restore it using the Restore Local Coprocessor Hard Drive Data procedure in the U^XS System Administration Manual. If not, “manually” configure user files. This can be done by using <code>ftp</code> to copy user files from other U^XSs (if available) , or by following the procedures in the Coprocessor User Environment Configuration section of the U^XS System Administration Manual.
3	Install any optional software by following the instructions that came with that software.
4	Perform a complete backup of your system by following the Backup the Local Coprocessor Hard Drive procedure in the U^XS System Administration Manual.

7.14 System Console Connection

Requirements for coprocessor console

Some system administration procedures will require a system console to complete. Examples of procedures requiring a system console are: restoring the software on a disk whose file system is corrupt, putting the U^XS in single user mode and performing a backup of the hard drive using the fbackup command, and troubleshooting more serious problems.

Recommended terminal

Honeywell recommends the Digital Equipment Corporation (DEC) VT-100 terminal (or any simple terminal with equivalent functionality) for use as a system console connected to the coprocessor.

You may also use a PC running suitable terminal emulation software (such as Windows 3.1 Terminal) to provide DEC VT-100 compatible terminal functionality.

A U^XS, or System Administration and Development workstation (Hewlett Packard), can also be used as a coprocessor console. The serial port of these devices must be connected to the serial port of your U^XS using the terminal cable to support this functionality.

Required terminal options

The terminal must be set to provide the following functionality to successfully communicate with the coprocessor serial port 1:

- Asynchronous operation
- 9600 bits per second
- 8 bits/character
- No parity

Overview

A direct connect interface cable must be connected between the system console device and the WSI2 I/O board (Serial Port 1).

Figure 7-1 illustrates that several different devices can be used to provide this system console connection. It also illustrates how the system console is connected to the coprocessor.

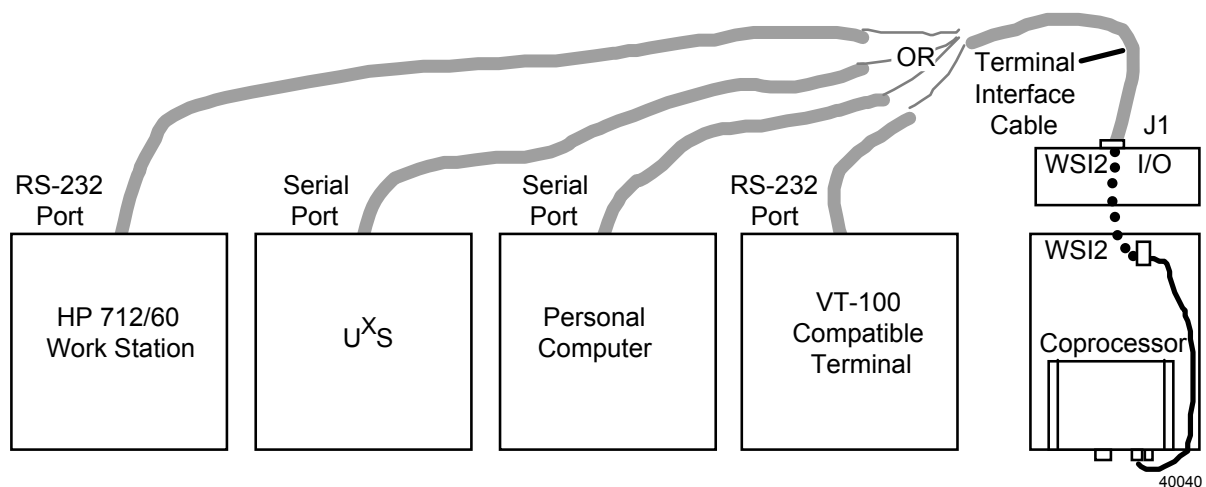


Figure 7-1 Coprocessor Console Connection

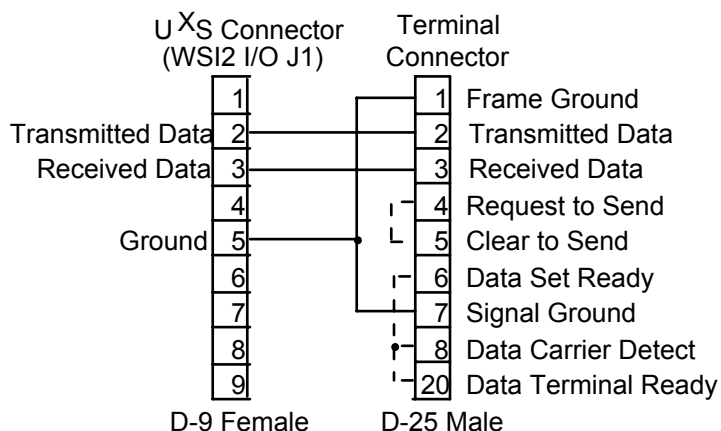
Coprocessor system console interface cable schematic

Examples of the direct connect terminal interface cable wiring details are shown in the following diagrams.

Notice that two cable types are described. One is a 9 to 25 pin direct connect cable and the other is a 9 to 9 pin direct connect cable.

The 9 to 25 pin cable (Figure 7-2) is available from Honeywell logistics using the part number 51196218-300. This is an optional cable. It is not generally supplied with the U^XS.

The 9 to 9 pin cable (Figure 7-3) is available from Honeywell logistics using the part number 51305069-100. This is also an optional cable. It is not generally supplied with the U^XS.



Notes:

1. Dashed connections may or may not be required by your terminal.
2. Connections on pins 2 and 3 may need to be reversed at one end and/or the D-25 may need to be a female connector, depending on the characteristics of your terminal.

40041

Figure 7-2 Coprocessor System Console Cable (9 to 25 pin)

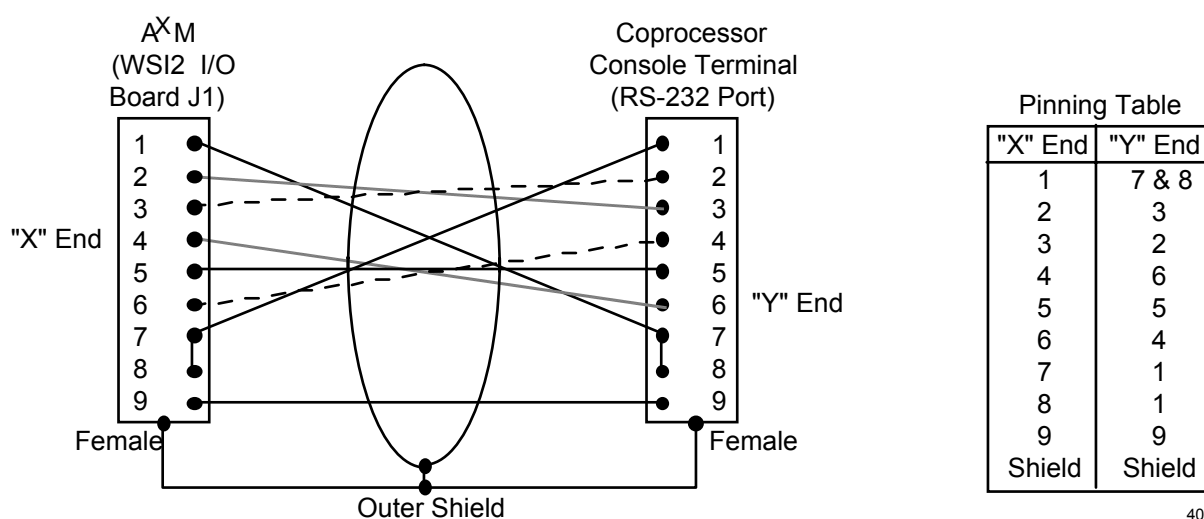


Figure 7-3 Coprocessor System Console Cable (9 to 9 pin)

⇒ **Note:** The dashed lines shown in the 9 to 9 cable diagram above are required connections. The dashed notation for some of the connections is used to make the specific wiring easier to read on screen.

Coproprocessor system console connection procedure

The following table defines the different cable types available for connecting coprocessor consoles. Note that these cables are not generally supplied with the U^XS. They are available by separate order.

Terminal Device	Terminal Connector	Cable Part Number
HP 712/60 Workstation	RS-232	51305069-100 9 pin(F) to 9 pin(F)
U ^X S	WSI I/O board (J1)	51305069-100 9 pin(F) to 9 pin(F)
VT-100 compatible devices with 9 pin RS-232 connector	Serial port	51305069-100 9 pin(F) to 9 pin(F)
VT-100 compatible devices with 25 pin RS-232 connector	Serial port	51196218-300 9 pin(F) to 25 pin(M)

Procedure overview

As mentioned previously, you may use any of the following devices for a system console:

- DEC VT-100 terminal (or any “dumb” terminal with equivalent functionality)
- PC running suitable terminal emulation software (such as Windows 3.1 Terminal) to provide DEC VT-100 compatible terminal functionality
- R300 U^XS
- HP workstation.

There are four procedures that follow this overview—one for each type of device. Make sure you use the correct procedure for the device you are using as a system console.

Dumb terminal procedure

Use the following procedure if you are using a “dumb terminal” as the system console.

Setting up VT-100 Emulation on a Dumb Terminal

⇒ **Note:** The coprocessor software defaults to the proper configuration of Serial Port 1 and no coprocessor configuration actions are required.

Step	Action
1	Connect the cable to the 25 pin RS-232 port on the terminal.
2	Connect the cable to the WSI2 I/O board connector for Serial Port 1 (connector J1).
3	Make sure the terminal is powered on.
4	Refer to the terminal documentation and set up the terminal for VT-100 emulation.
5	Set up the following communication parameters: 9600 baud, 8 data bits, 1 stop bit, no parity
6	If the U ^X S is not powered on, turn power on now (you can watch the bootup process) Note: Be sure that the toggle switch on the left side of the WSI2 board is in the right (RUN) position.
7	At your system console, press <RETURN>. You should receive the login prompt from the U ^X S
8	Log in as root.
9	Enter <code>setenv TERM vt100</code> and press <RETURN>.
10	Perform the desired procedure (restore software, backup, etc.). After the procedure is complete, return to the next step in this procedure.
11	Enter <code>exit</code> and press <RETURN>.

PC procedure

Use the following procedure if you are using a PC with Windows 3.1 for the system console.

Starting VT-100 Emulation on a PC.

⇒ **Note:** The coprocessor software defaults to the proper configuration of Serial Port 1 and no coprocessor configuration actions are required.

Step	Action
1	Connect the cable to the RS-232 port on the back of the PC. Use the end marked "HOST RS-232 PORT". Note: Henceforth, the term system console means the PC.
2	Connect the cable to the WSI2 I/O board connector for Serial Port 1 (connector J1). Use the end marked "WSI2 I/O J1".
3	Start Windows 3.1 and (if necessary) login to the PC.
4	At the Program Manager, double-click on the Accessories icon.
5	Double-click on the Terminal icon Note: If you get the Terminal Error "The selected COM port is either not supported or is being used by another device. Select another port.", Click OK. When the Communications window is displayed, Click Cancel.
6	Pull down the Settings menu and select Terminal Emulation – Select DEC VT-100 (ANSI) – Click OK
7	Pull down the Settings menu and select Terminal Preferences • Under CR->CR/LF, ensure neither box is selected • Under Columns, ensure 80 is selected • At the bottom, ensure Show Scroll Bars is not selected and ensure Use Function, Arrow, and Ctrl Keys for Windows is not selected (very important) • Click OK
8	Pull down the Settings menu and select Communications – Select COM1 – Select 9600 – Select 8 data bits – Select 1 stop bit – Select Parity None – Select Flow Control None – Parity Check and Carrier Detect should not be selected – Click OK
9	Maximize the window.
10	If you wish to save your configuration settings, pull down the File menu, select Save As, enter a filename, and click OK. The next time you enter the terminal emulation program, pull down the File menu and select Open. This will automatically restore your settings.
11	If the U ^X S is not powered on, turn power on now (you can watch the bootup process). Note: Be sure that the toggle switch on the left side of the WSI2 board is in the right (RUN) position.

Coprocessor Administration Tasks – System Console Connection

Step	Action
12	At your system console, press <RETURN>. You should receive the login prompt from the U ^x S.
13	Log in as root.
14	Enter <code>setenv TERM vt100</code> and press <RETURN>.
15	Perform the desired procedure (restore software, backup, etc.). After the procedure is complete, return to the next step in this procedure.
16	Enter <code>exit</code> and press <RETURN>.
17	Close the system terminal window.

U^XS procedure (R300)

Use the following procedure if you are using an R300 U^XS to serve as the system console. The console serial port (J1) is used on both the “terminal” U^XS, and the U^XS to which a terminal is being connected.

Starting VT-100 Emulation on an R300 U^XS

Step	Action
1	Connect the cable to the WSI2 I/O board connector for Serial Port 1 (connector J1) on the “terminal” U ^X S. Use the end marked “HOST RS-232 PORT”. Note: Henceforth, the term system console means the “terminal” U ^X S.
2	Connect the cable to the WSI2 I/O board connector for Serial Port 1 (connector J1). Use the end marked “WSI2 I/O J1”.
3	At the “terminal” U ^X S, login as engineer.
4	Select Application Menu->Telnet from the workspace menu.
5	Enter <code>!</code> and press <RETURN>.
6	Set up the backspace key. Enter <code>stty erase <SPACE> <BACKSPACE></code> and press <RETURN>.
7	Become the root user. Enter <code>su -root</code> and press <RETURN>.
8	Enter the coprocessor (root) password and press <RETURN>.
9	Find the Process Identification Number (PID) of the console process. Enter <code>ps -ef grep console</code> and press <RETURN>.
10	You should see a line of the form: <code>root nnn...../etc/getty -h console console</code> <i>nnn</i> is the PID—record it here _____
11	Enter <code>kill -STOP nnn</code> and press <RETURN>. using the PID for <i>nnn</i> .
12	Invoke Kermit. Enter <code>kermit -l /dev/console -b 9600 -c</code> and press <RETURN>. Note: <code>l</code> is the letter ell
13	If the U ^X S is not powered on, turn power on now (you can watch the bootup process) Note: Be sure that the toggle switch on the left side of the WSI2 board is in the right (RUN) position.
14	At your system console, press <RETURN>. You should receive the login prompt from the U ^X S
15	Log in as root.

Coprocessor Administration Tasks – System Console Connection

Step	Action
16	Enter <code>setenv TERM xterm</code> and press <RETURN>.
17	Perform the desired procedure (restore software, backup, etc.). After the procedure is complete, return to the next step in this procedure.
18	Enter <code>exit</code> and press <RETURN>.
19	Close the system console window. Note: It is important to close this window even though you are going to open another xterm window.
20	Open an xterm window and become the root user by repeating steps 4-8 of this procedure.
21	Remove the “exclusive use of the system console port (J1) lock”. Enter <code>rm /usr/spool/uucp/LCK..console</code> and press <RETURN>.
22	Enter <code>kill -CONT nnn</code> and press <RETURN>. using the PID recorded earlier for <i>nnn</i> .
23	Close the xterm window.
24	Logout.

Hewlett Packard workstation procedure

Use the following procedure if you are using the serial port on the Honeywell MP-AMXST1 (HP workstation) for the system console.

Starting VT-100 Emulation on a HP workstation

⇒ **Note:** The coprocessor software defaults to the proper configuration of Serial Port 1 and no coprocessor configuration actions are required.

Step	Action
1	Connect the cable to the RS-232 port on the back of the HP workstation. Use the end marked "HOST RS-232 PORT". Note: Henceforth, the term system console means the HP workstation
2	Connect the cable to the WSI2 I/O board connector for Serial Port 1 (connector J1). Use the end marked "WSI2 I/O J1".
3	Login to the HP workstation.
4	Open an hp terminal window by clicking on the terminal icon.
5	At the "TERM = (hp)" prompt, press <RETURN>.
6	Enter <code>kermit</code> and press <RETURN>.
7	Enter <code>set line /dev/tty00</code> and press <RETURN>.
8	Enter <code>set baud 9600</code> and press <RETURN>.
9	Enter <code>log session pathname</code> and press <RETURN>. where <i>pathname</i> is a file path of your choice for a session log file (for example, <code>/tmp/session.log</code>) Note: Do not skip this step, even if you don't need a log. On the HP workstation, Kermit has a tendency to hang if the session is not logged.
10	Enter <code>connect</code> and press <RETURN>.
11	If the U ^X S is not powered on, turn power on now (you can watch the bootup process). Note: Be sure that the toggle switch on the left side of the WSI2 board is in the right (RUN) position.
12	At the system console, press <RETURN>. You should receive the login prompt from the U ^X S
13	Log in as root.

Coprocessor Administration Tasks – System Console Connection

Step	Action
14	Perform the desired procedure (restore software, backup, etc.). After the procedure is complete, return to the next step in this procedure.
15	Enter <code>exit</code> and press <RETURN>.
16	Close the system console window and Logout.

7.15 Change Logical Volume Sizes

Overview

LVM, which is a disk management subsystem, lets you allocate disk space according to the specific or projected sizes of your file systems or raw data. If your hard drive is configured with LVM, you can use SAM to change the volume sizes, and also to check the size of current volumes.

However, there may be times when you are unable to use SAM to change the size of logical volumes because you may have to unmount the filesystem in which SAM resides.

Alternatively, you may also use the command line method to change logical volume sizes.

If you do change the size of your logical volumes, you should create an updated bootable system recovery tape so that if it becomes necessary to use the tape for system recovery, the most current logical volume sizes will be restored.

Procedure

Step	Action
1	Determine the logical volume name of the filesystem you want to increase. For example, <code>/dev/vg00/lvol8</code> <code>bdf</code>
2	Go to single user mode. <code>/sbin/shutdown now</code>
3	Unmount the filesystem. <code>/sbin/umount [filesystem] (ie. /usr)</code>
4	Extend the logical volume. <code>/sbin/lvextend -L XXX [logical volume name]</code> Where: • XXX = the size in mb that you want the volume to be. • Logical volume name = the name obtained in step 1 above. For example, <code>/dev/vg00/lvol8</code>
5	Extend the filesystem. <code>/sbin/extendfs [raw logical device]</code> For example, raw logical device = <code>/dev/vg00/r1vol8</code>

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